

UMC 0.11um RFCMOS AI Advanced Enhancement Standard Performance 1.2V SPICE Model Document

Version 0.2 phase1

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1 Introduction

1.1 Revision history

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2011/02/11	Chih-Yang Kao	1. Original release 2. Silicon-based model

1.2 General Information

Following Logic/MM model is used as the main body of the subckt RF MOSFET model and it would be fine tuned in order to improve Gm and Rout characteristics.

l110ae_sp12_v021_rf

More information about the model is specified below.

Model File Name:

Spectre: l110ae_sp12_v021_rf.lib.scs, l110ae_sp12_v021_rf.mdl.scs

Hspice: l110ae_sp12_v021_rf.lib, l110ae_sp12_v021_rf.mdl

Eldo: l110ae_sp12_v021_rf.lib.eldo, l110ae_sp12_v021_rf.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

Twin Well NMOS: n_12_sprf (drain gate source body)

PMOS: p_12_sprf (drain gate source body psub)

Triple Well NMOS: n_12bpw_sprf (drain gate source body nwell psub)

Corner Section:

tt : Typical n-ch MOSFET and Typical p-ch MOSFET

ff : Fast n-ch MOSFET and Fast p-ch MOSFET

ss : Slow n-ch MOSFET and Slow p-ch MOSFET

fnsf : Fast n-ch MOSFET and Slow p-ch MOSFET

snfp : Slow n-ch MOSFET and Fast p-ch MOSFET

Input Parameters:

lf : gate finger length in m

wf : gate finger width in m

nf : gate finger number

Geometry scaling:

NMOS:

Finger Length (lf): 0.12um ~ 0.36um

Finger Width (wf): 0.9um ~ 7.2um

Finger No (nf): 2~ 22

Support even and odd finger number

PMOS:

Finger Length (lf): 0.12um ~ 0.36um

Finger Width (wf): 1.6um ~ 9.6um

Finger No (nf): 2 ~ 22

Support even and odd finger number

Triple-well NMOS:

Finger Length (lf): 0.12um ~ 0.36um

Finger Width (wf): 0.9um ~ 7.2um

Finger No (nf): 2~ 22

Support even and odd finger number

Biasing coverage:

Voltage: Upto Vdd=1.2V

Frequency: upto 20.1GHz

Temperature coverage:

-50 C ~ +125 C

Simulator Versions:

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 6.2.1.299

Mentor Graphics ELDO version 6.11_1.1

Wafer Information:

Table 1-2 Wafer Information

Mask ID	BSW0FA
Lot ID	D54AH
Wafer Number	23

1.3 Model Usage:

1.3.1 For Cadence Spectre

- Copy ModelFileName.lib.scs and ModelFileName.mdl.scs into your simulation directory.
- Add the following statement in the input netlist to the simulator
include "./ModelFileName.lib.scs " section = CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
- SetOption1 options scale=0.9
- Follow the example below to call the model
x1 (drain gate source body) n_12_sprf lf=0.12e-6 wf=3.6e-6 nf=4 m=8

1.3.2 For Synopsis Hspice

- Copy ModelFileName.lib and ModelFileName.mdl into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_12_sprf lf=0.12e-6 wf=3.6e-6 nf=4 m=8

1.3.3 For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo and ModelFileName.mdl.eldo into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_12_sprf lf=0.12e-6 wf=3.6e-6 nf=4 m=8

ModelFileName: 1110ae_sp12_v021_rf

1.4 Device Architecture and Layout

1.4.1 Twin Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

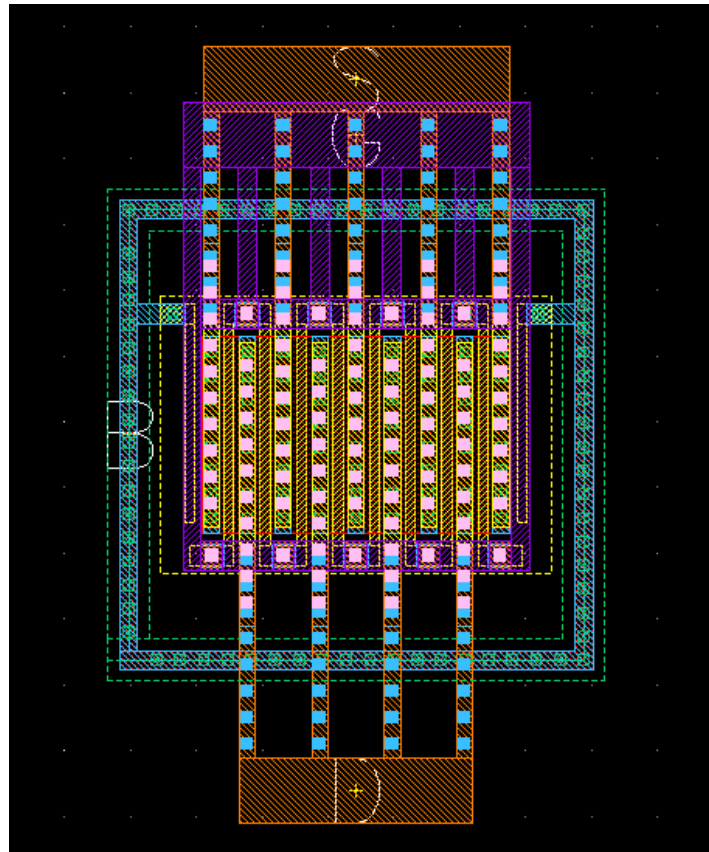


Figure 1-1 Example layout of 1.2V pwell RF NMOS

Extension equivalent circuit model:

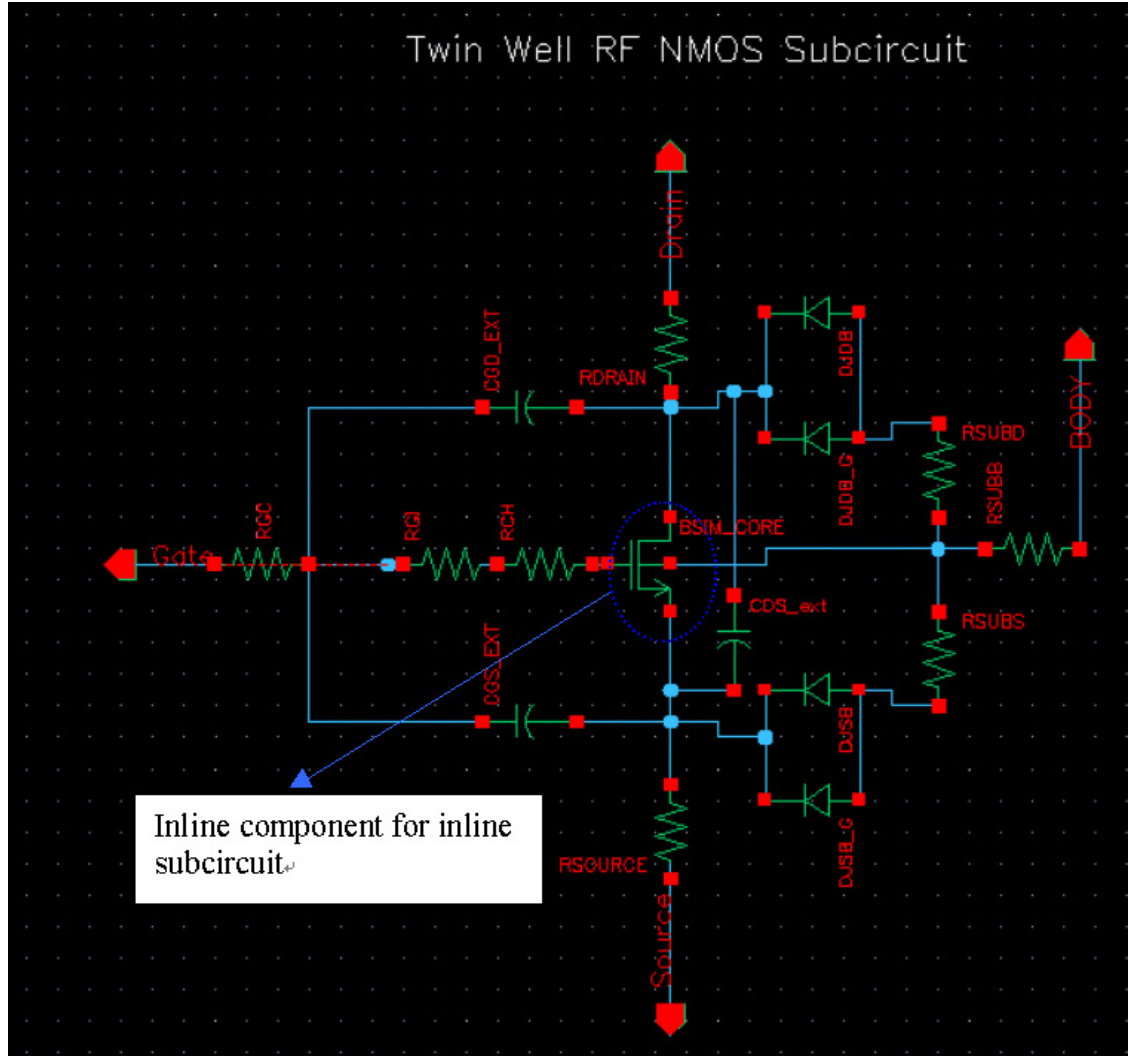


Figure 1-2 Extension equivalent circuit of TWIN-Well NMOS

1.4.2 PMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

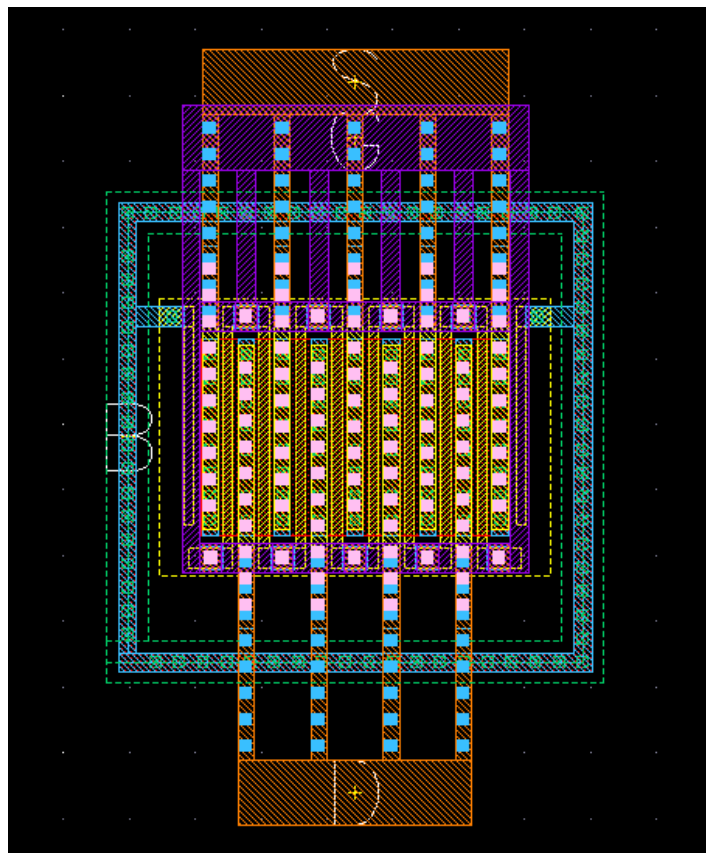


Figure 1-3 Example layout of 1.2V pwell RF NMOS

Extension equivalent circuit model:

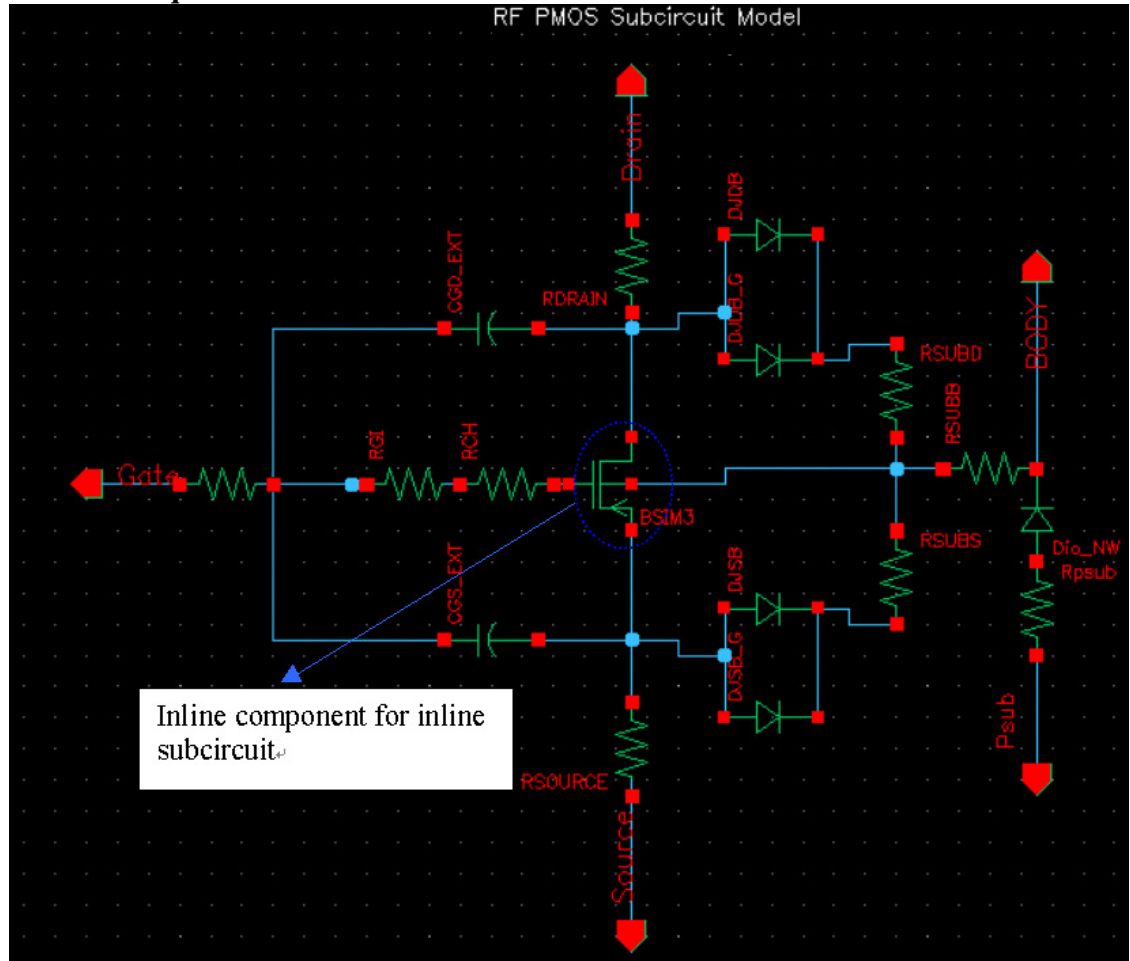


Figure 1-4 Extension equivalent circuit of PMOS

1.4.3 Triple Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.12um.

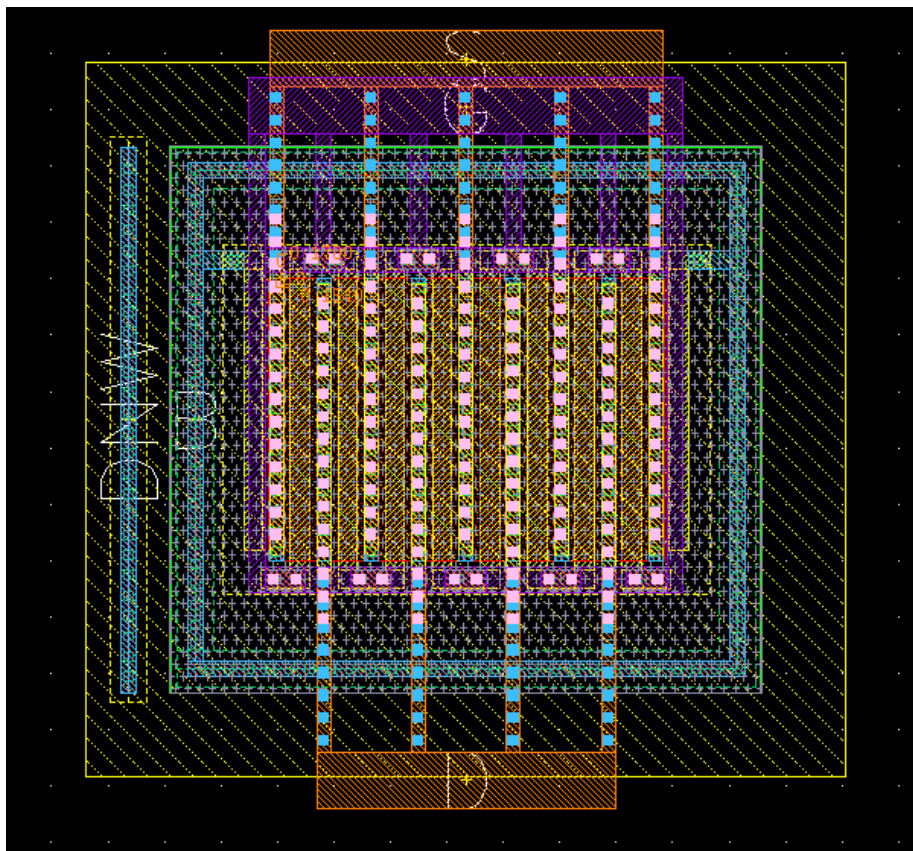


Figure 1-5 Example layout of Triple-Well RF NMOS

Extension equivalent circuit model:

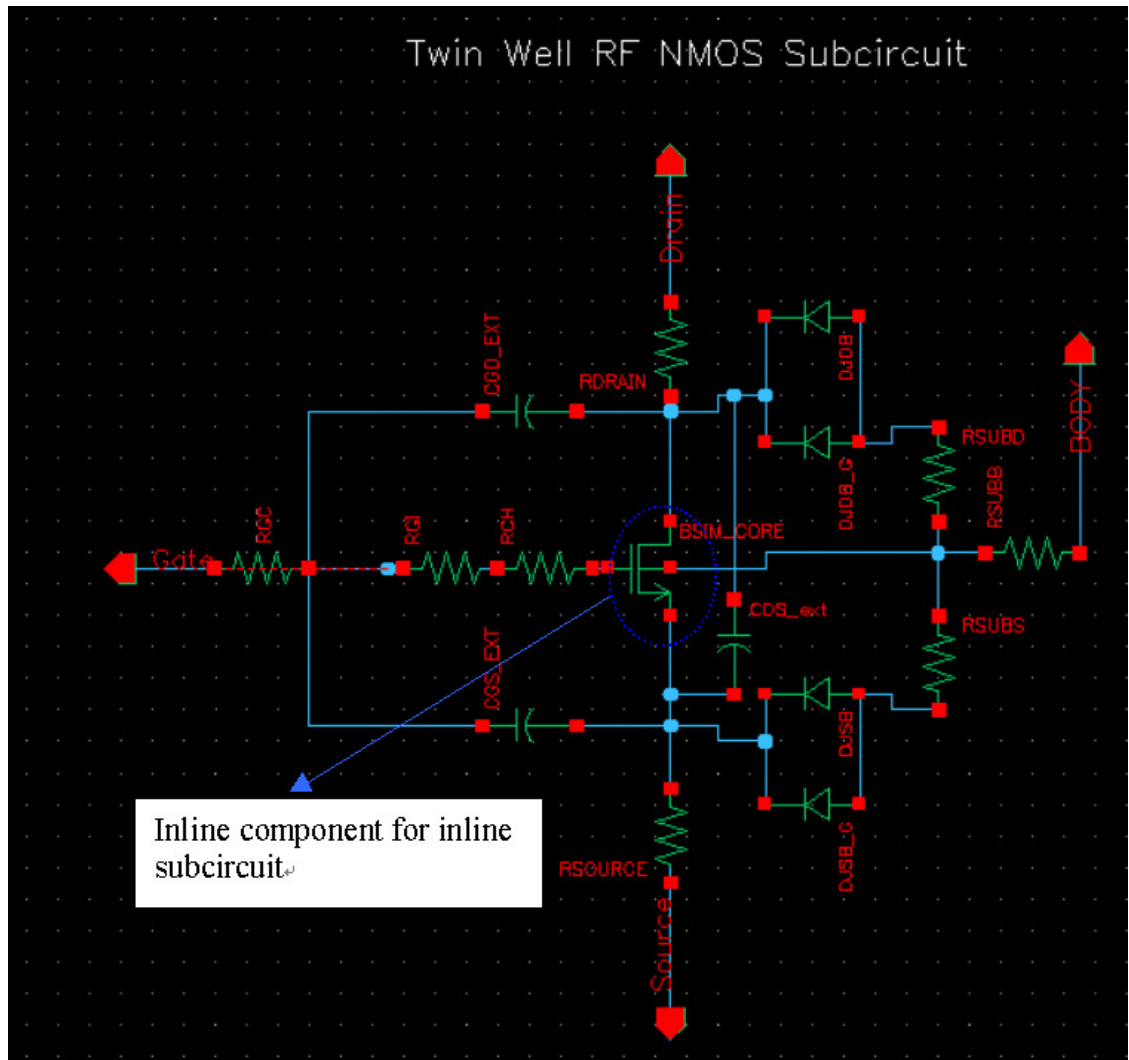


Figure 1-6 Sub-circuit topology of 1.2V Triple-Well RF NMOS

1.5 Model Features

BSIM3 :

The BSIM3 model in RF subcircuit model is based on Logic/MM model, with parameters refined to improve the fitting quality of G_m / R_{out} and RF performance.

LOD effect :

LOD model has been added to model STI stress & Finger number effects.

1/f Noise :

The BSIM3 1/f noise model is included

Note:

Model can support odd finger numbers, but it is recommended to use even finger number to reduce drain junction capacitance.

It is recommended not to use the model outside the above WF, LF and NF range. Outside above range, the model fitting quality may become worse.

This model should be used in conjunction with RF FET Pcell (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The model may not be valid if the transistor layout is different from UMC Pcell as shown in above layout example.

2 RF Model Verification

2.1 Twin Well NMOS Model

2.1.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max : Vd=1.2V, sweep Vg
vth : Linear Vth extracted @ Vd=0.05V @
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=1.2V using fixed current
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ -1.2V
Idsat : Drain current extracted @ Vd=Vg=1.2V
Idlin : Drain current extracted @ Vd=0.05V, Vg=1.2V
Ioff : Drain current extracted @ Vg=0V, Vd=1.2V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	0.36	0.12	0.12	0.12	0.36	0.12	0.12	0.12
	WF (um)	3.6	7.2	1.8	0.9	3.6	7.2	1.8	0.9
	NF	16	16	16	16	16	16	16	16
SS	ft(GHz)	20.444	95.122	70.569	52.463	NA	NA	NA	NA
	idsat(uA/um)	2.07E+02	4.15E+02	4.10E+02	4.19E+02	2.11E+02	4.56E+02	4.51E+02	4.42E+02
	vtlin(V)	4.46E-01	5.35E-01	5.56E-01	5.58E-01	4.43E-01	5.04E-01	5.16E-01	5.28E-01
	vtsat(V)	4.23E-01	4.45E-01	4.63E-01	4.66E-01	4.09E-01	4.16E-01	4.30E-01	4.45E-01
	body_effect(V)	1.95E-01	1.31E-01	1.32E-01	1.34E-01	1.88E-01	1.16E-01	1.14E-01	1.13E-01
	ioff(pA/um)	3.01E+00	1.39E+01	1.18E+01	1.52E+01	3.82E+00	2.64E+01	2.38E+01	2.20E+01
SNFP	ft(GHz)	20.949	97.398	79.287	63.035	NA	NA	NA	NA
	idsat(uA/um)	2.26E+02	4.58E+02	4.63E+02	4.80E+02	2.33E+02	5.05E+02	5.07E+02	5.06E+02
	vtlin(V)	4.19E-01	5.02E-01	5.16E-01	5.14E-01	4.16E-01	4.72E-01	4.77E-01	4.82E-01
	vtsat(V)	3.96E-01	4.12E-01	4.24E-01	4.22E-01	3.82E-01	3.83E-01	3.91E-01	3.99E-01
	body_effect(V)	1.89E-01	1.27E-01	1.29E-01	1.30E-01	1.84E-01	1.11E-01	1.10E-01	1.09E-01
	ioff(pA/um)	6.02E+00	2.95E+01	2.74E+01	3.69E+01	7.92E+00	6.68E+01	6.69E+01	6.81E+01
TT	ft(GHz)	22.579	106.77	90.294	73.399	NA	NA	NA	NA
	idsat(uA/um)	2.47E+02	5.04E+02	5.17E+02	5.38E+02	2.54E+02	5.54E+02	5.61E+02	5.67E+02
	vtlin(V)	3.92E-01	4.70E-01	4.79E-01	4.70E-01	3.89E-01	4.39E-01	4.40E-01	4.40E-01
	vtsat(V)	3.70E-01	3.81E-01	3.86E-01	3.77E-01	3.56E-01	3.51E-01	3.53E-01	3.56E-01
	body_effect(V)	1.89E-01	1.13E-01	1.17E-01	1.21E-01	1.84E-01	1.10E-01	1.09E-01	1.08E-01
	ioff(pA/um)	1.27E+01	1.02E+02	1.03E+02	1.57E+02	1.80E+01	1.82E+02	2.01E+02	2.26E+02
FNFP	ft(GHz)	23.91	117.52	103.64	87.064	NA	NA	NA	NA
	idsat(uA/um)	2.68E+02	5.49E+02	5.72E+02	5.98E+02	2.77E+02	6.03E+02	6.15E+02	6.26E+02
	vtlin(V)	3.66E-01	4.37E-01	4.42E-01	4.30E-01	3.63E-01	4.07E-01	4.03E-01	3.98E-01
	vtsat(V)	3.44E-01	3.50E-01	3.50E-01	3.36E-01	3.30E-01	3.18E-01	3.16E-01	3.14E-01
	body_effect(V)	1.89E-01	9.42E-02	1.02E-01	1.09E-01	1.84E-01	1.09E-01	1.08E-01	1.07E-01
	ioff(pA/um)	2.82E+01	3.59E+02	3.85E+02	5.98E+02	4.26E+01	4.77E+02	5.85E+02	7.32E+02
FF	ft(GHz)	24.423	119.21	117.16	109.64	NA	NA	NA	NA
	idsat(uA/um)	2.90E+02	5.93E+02	6.25E+02	6.56E+02	3.00E+02	6.52E+02	6.69E+02	6.86E+02
	vtlin(V)	3.39E-01	4.05E-01	4.04E-01	3.88E-01	3.36E-01	3.74E-01	3.66E-01	3.56E-01
	vtsat(V)	3.17E-01	3.17E-01	3.12E-01	2.94E-01	3.03E-01	2.87E-01	2.81E-01	2.74E-01
	body_effect(V)	1.83E-01	9.22E-02	9.94E-02	1.06E-01	1.80E-01	1.06E-01	1.05E-01	1.05E-01
	ioff(pA/um)	6.49E+01	7.14E+02	8.81E+02	1.57E+03	1.01E+02	1.12E+03	1.53E+03	2.14E+03

Table 2-1 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	0.36	0.12	0.12	0.12	0.36	0.12	0.12	0.12
	WF (um)	3.6	7.2	1.8	0.9	3.6	7.2	1.8	0.9
	NF	16	16	16	16	16	16	16	16
SS	ft(GHz)	18.203	85.697	63.655	47.309	NA	NA	NA	NA
	idsat(uA/um)	1.88E+02	3.82E+02	3.79E+02	3.91E+02	1.85E+02	4.35E+02	4.22E+02	4.07E+02
	vtlin(V)	3.81E-01	4.76E-01	4.97E-01	4.99E-01	3.76E-01	4.45E-01	4.59E-01	4.73E-01
	vtsat(V)	3.51E-01	3.79E-01	3.97E-01	4.00E-01	3.34E-01	3.46E-01	3.62E-01	3.79E-01
	body_effect(V)	2.08E-01	1.61E-01	1.61E-01	1.60E-01	2.00E-01	1.28E-01	1.26E-01	1.24E-01
	ioff(pA/um)	2.80E+02	1.18E+03	9.28E+02	1.03E+03	4.00E+02	2.09E+03	1.77E+03	1.48E+03
SNFP	ft(GHz)	18.8	88.169	71.82	57.008	NA	NA	NA	NA
	idsat(uA/um)	2.04E+02	4.21E+02	4.26E+02	4.44E+02	2.04E+02	4.83E+02	4.76E+02	4.68E+02
	vtlin(V)	3.54E-01	4.42E-01	4.56E-01	4.54E-01	3.48E-01	4.11E-01	4.19E-01	4.26E-01
	vtsat(V)	3.25E-01	3.45E-01	3.57E-01	3.55E-01	3.08E-01	3.13E-01	3.22E-01	3.32E-01
	body_effect(V)	2.03E-01	1.57E-01	1.56E-01	1.55E-01	1.97E-01	1.23E-01	1.22E-01	1.21E-01
	ioff(pA/um)	5.25E+02	2.16E+03	1.91E+03	2.29E+03	7.63E+02	4.27E+03	4.01E+03	3.78E+03
TT	ft(GHz)	20.283	94.261	81.047	65.992	NA	NA	NA	NA
	idsat(uA/um)	2.22E+02	4.64E+02	4.75E+02	4.96E+02	2.21E+02	5.31E+02	5.27E+02	5.24E+02
	vtlin(V)	3.27E-01	4.11E-01	4.20E-01	4.11E-01	3.22E-01	3.79E-01	3.81E-01	3.83E-01
	vtsat(V)	2.98E-01	3.14E-01	3.20E-01	3.10E-01	2.81E-01	2.80E-01	2.84E-01	2.89E-01
	body_effect(V)	2.03E-01	1.44E-01	1.46E-01	1.47E-01	1.97E-01	1.21E-01	1.20E-01	1.20E-01
	ioff(pA/um)	1.01E+03	5.30E+03	5.15E+03	6.91E+03	1.47E+03	8.97E+03	9.19E+03	9.48E+03
FNFP	ft(GHz)	21.496	101.67	90.534	76.138	NA	NA	NA	NA
	idsat(uA/um)	2.40E+02	5.06E+02	5.24E+02	5.47E+02	2.39E+02	5.78E+02	5.78E+02	5.80E+02
	vtlin(V)	2.99E-01	3.79E-01	3.84E-01	3.71E-01	2.95E-01	3.46E-01	3.44E-01	3.41E-01
	vtsat(V)	2.71E-01	2.84E-01	2.84E-01	2.69E-01	2.55E-01	2.48E-01	2.47E-01	2.46E-01
	body_effect(V)	2.03E-01	1.27E-01	1.33E-01	1.37E-01	1.97E-01	1.21E-01	1.20E-01	1.19E-01
	ioff(pA/um)	1.93E+03	1.27E+04	1.32E+04	1.82E+04	2.83E+03	1.84E+04	2.04E+04	2.29E+04
FF	ft(GHz)	22.22	102.73	101.53	94.891	NA	NA	NA	NA
	idsat(uA/um)	2.60E+02	5.49E+02	5.73E+02	6.00E+02	2.59E+02	6.26E+02	6.30E+02	6.38E+02
	vtlin(V)	2.72E-01	3.45E-01	3.45E-01	3.28E-01	2.67E-01	3.13E-01	3.06E-01	2.98E-01
	vtsat(V)	2.44E-01	2.50E-01	2.45E-01	2.26E-01	2.28E-01	2.16E-01	2.11E-01	2.05E-01
	body_effect(V)	1.98E-01	1.24E-01	1.29E-01	1.34E-01	1.93E-01	1.18E-01	1.18E-01	1.17E-01
	ioff(pA/um)	3.65E+03	2.23E+04	2.54E+04	3.85E+04	5.41E+03	3.51E+04	4.21E+04	5.14E+04

Table 2-2 RF & Logic Corner Table at 125 °C

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	0.36	0.12	0.12	0.12	0.36	0.12	0.12	0.12
	WF (um)	3.6	7.2	1.8	0.9	3.6	7.2	1.8	0.9
	NF	16	16	16	16	16	16	16	16
SS	ft(GHz)	22.719	107.57	79.286	58.818	NA	NA	NA	NA
	idsat(uA/um)	2.28E+02	4.47E+02	4.42E+02	4.51E+02	2.46E+02	4.75E+02	4.78E+02	4.75E+02
	vtlin(V)	5.05E-01	5.91E-01	6.11E-01	6.13E-01	5.02E-01	5.60E-01	5.70E-01	5.80E-01
	vtsat(V)	4.85E-01	5.03E-01	5.21E-01	5.23E-01	4.74E-01	4.77E-01	4.89E-01	5.03E-01
	body_effect(V)	1.86E-01	1.03E-01	1.07E-01	1.10E-01	1.78E-01	1.08E-01	1.06E-01	1.04E-01
	ioff(pA/um)	8.90E-01	5.27E-01	1.70E+00	3.29E+00	7.45E-01	4.42E-01	1.54E+00	3.02E+00
SNFP	ft(GHz)	23.116	110.16	89.325	70.943	NA	NA	NA	NA
	idsat(uA/um)	2.49E+02	4.95E+02	5.03E+02	5.21E+02	2.71E+02	5.24E+02	5.34E+02	5.40E+02
	vtlin(V)	4.78E-01	5.58E-01	5.71E-01	5.69E-01	4.76E-01	5.28E-01	5.32E-01	5.35E-01
	vtsat(V)	4.59E-01	4.71E-01	4.82E-01	4.80E-01	4.48E-01	4.45E-01	4.51E-01	4.58E-01
	body_effect(V)	1.80E-01	1.01E-01	1.04E-01	1.07E-01	1.75E-01	1.03E-01	1.01E-01	1.00E-01
	ioff(pA/um)	1.13E+00	8.16E-01	1.99E+00	3.60E+00	7.54E-01	6.15E-01	1.74E+00	3.24E+00
TT	ft(GHz)	24.815	121.95	103.01	83.816	NA	NA	NA	NA
	idsat(uA/um)	2.73E+02	5.43E+02	5.62E+02	5.86E+02	2.97E+02	5.72E+02	5.89E+02	6.02E+02
	vtlin(V)	4.52E-01	5.24E-01	5.33E-01	5.24E-01	4.50E-01	4.96E-01	4.94E-01	4.92E-01
	vtsat(V)	4.33E-01	4.37E-01	4.43E-01	4.33E-01	4.22E-01	4.12E-01	4.13E-01	4.15E-01
	body_effect(V)	1.80E-01	8.52E-02	9.06E-02	9.67E-02	1.74E-01	1.01E-01	1.00E-01	9.89E-02
	ioff(pA/um)	1.15E+00	1.25E+00	2.45E+00	4.46E+00	7.84E-01	1.31E+00	2.61E+00	4.37E+00
FNFP	ft(GHz)	26.288	135.16	119.54	100.7	NA	NA	NA	NA
	idsat(uA/um)	2.98E+02	5.89E+02	6.22E+02	6.53E+02	3.23E+02	6.21E+02	6.43E+02	6.62E+02
	vtlin(V)	4.26E-01	4.90E-01	4.94E-01	4.82E-01	4.24E-01	4.63E-01	4.58E-01	4.51E-01
	vtsat(V)	4.07E-01	4.04E-01	4.04E-01	3.91E-01	3.96E-01	3.80E-01	3.77E-01	3.73E-01
	body_effect(V)	1.79E-01	6.56E-02	7.44E-02	8.32E-02	1.75E-01	1.00E-01	9.94E-02	9.84E-02
	ioff(pA/um)	1.21E+00	3.81E+00	5.36E+00	9.79E+00	8.81E-01	3.77E+00	6.20E+00	9.76E+00
FF	ft(GHz)	26.809	137.3	135.55	127.41	NA	NA	NA	NA
	idsat(uA/um)	3.21E+02	6.34E+02	6.79E+02	7.16E+02	3.50E+02	6.69E+02	6.97E+02	7.22E+02
	vtlin(V)	3.99E-01	4.59E-01	4.58E-01	4.41E-01	3.97E-01	4.32E-01	4.22E-01	4.10E-01
	vtsat(V)	3.80E-01	3.73E-01	3.68E-01	3.50E-01	3.70E-01	3.49E-01	3.42E-01	3.33E-01
	body_effect(V)	1.73E-01	6.40E-02	7.26E-02	8.13E-02	1.71E-01	9.77E-02	9.69E-02	9.61E-02
	ioff(pA/um)	2.19E+00	8.74E+00	1.26E+01	2.62E+01	1.19E+00	1.08E+01	1.82E+01	3.10E+01

Table 2-3 RF & Logic Corner Table at -50 °C

2.1.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
NFET551	0.12	1.8	16	4	1	0.482	0.479	0.372	0.386	516.51	503.39	150.6	103.15	89.643	85.523
NFET554	0.12	7.2	16	1	1	0.481	0.47	0.371	0.381	468.92	478.61	144.1	102.31	100.906	100.64
NFET555	0.12	0.9	16	6	1	0.476	0.47	0.378	0.377	517.34	530.83	102.76	157.22	70.507	71.353
NFET556	0.12	1.8	2	18	1	0.491	0.479	0.385	0.386	502.33	512.31	109.25	103.8	79.465	80.261
NFET563	0.15	2.4	16	3	1	0.466	0.464	0.405	0.406	416.07	406.61	23.47	15.55	67.29	65.712
NFET552	0.12	2.4	16	3	1	0.484	0.478	0.374	0.386	504.73	500.42	139	98.597	93.548	92.306
NFET553	0.12	3.6	16	2	1	0.483	0.475	0.373	0.384	498.09	492.56	145.06	97.899	99.319	96.87
NFET558	0.12	1.8	5	13	1	0.489	0.479	0.38	0.386	502.15	509.57	125.19	103.35	84.476	85.465
NFET559	0.12	1.8	8	8	1	0.487	0.479	0.376	0.386	508.85	508.48	137.89	103.24	87.311	86.588
NFET560	0.12	1.8	11	6	1	0.486	0.479	0.377	0.386	509.47	507.8	124.13	103.19	88.867	87.368
NFET561	0.12	1.8	22	3	1	0.483	0.479	0.375	0.386	510.48	503.27	137.73	103.12	89.85	86.427
NFET564	0.18	2.4	16	3	1	0.442	0.446	0.395	0.405	373.82	357.99	20.46	10.693	51.893	50.63
NFET565	0.24	3.6	16	2	1	0.421	0.421	0.384	0.392	314.66	300.58	16.51	9.787	37.137	36.799
NFET566	0.36	3.6	16	2	1	0.397	0.392	0.367	0.37	240.57	242.72	16.95	12.669	20.904	22.245
NFET570	0.12	3.6	4	8	1	0.488	0.475	0.38	0.384	490.8	498.76	116.65	98.038	94.447	95.514
NFET571	0.12	3.6	11	4	1	0.486	0.475	0.378	0.384	488.79	493.3	116.16	97.92	96.383	96.457
NFET572	0.12	3.6	22	2	1	0.483	0.475	0.375	0.384	492.79	490.08	132.49	97.887	97.058	95.932
NFET573	0.15	7.2	4	4	1	0.476	0.462	0.415	0.403	380.67	397.3	17.12	13.002	64.83	69.16
NFET576	0.12	1.8	4	1	1	0.491	0.479	0.386	0.386	511.14	512.83	101.32	103.43	100.966	85.365

Table 2-4 Summary Table for Twin Well RF NMOS

Measure: Measurement result.

Model: Simulation result

2.1.3 Simulation results for WF, LF and NF trends

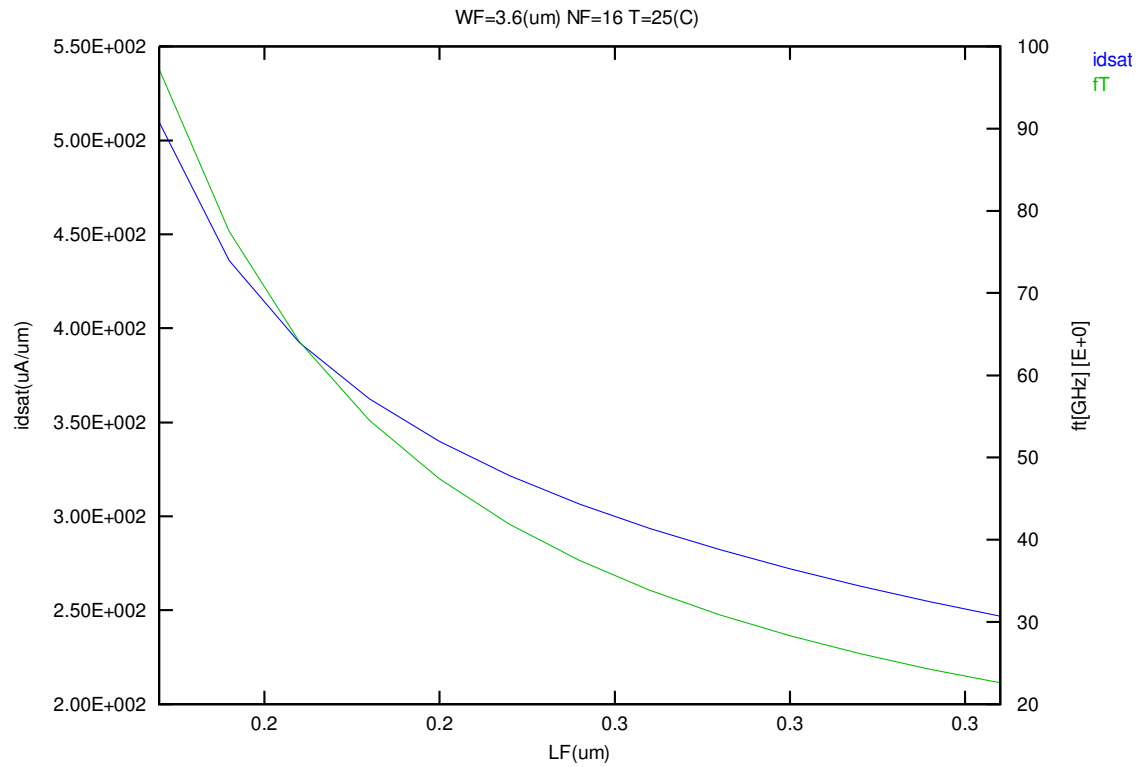


Figure 2-1 fT and Idsat vs. channel length

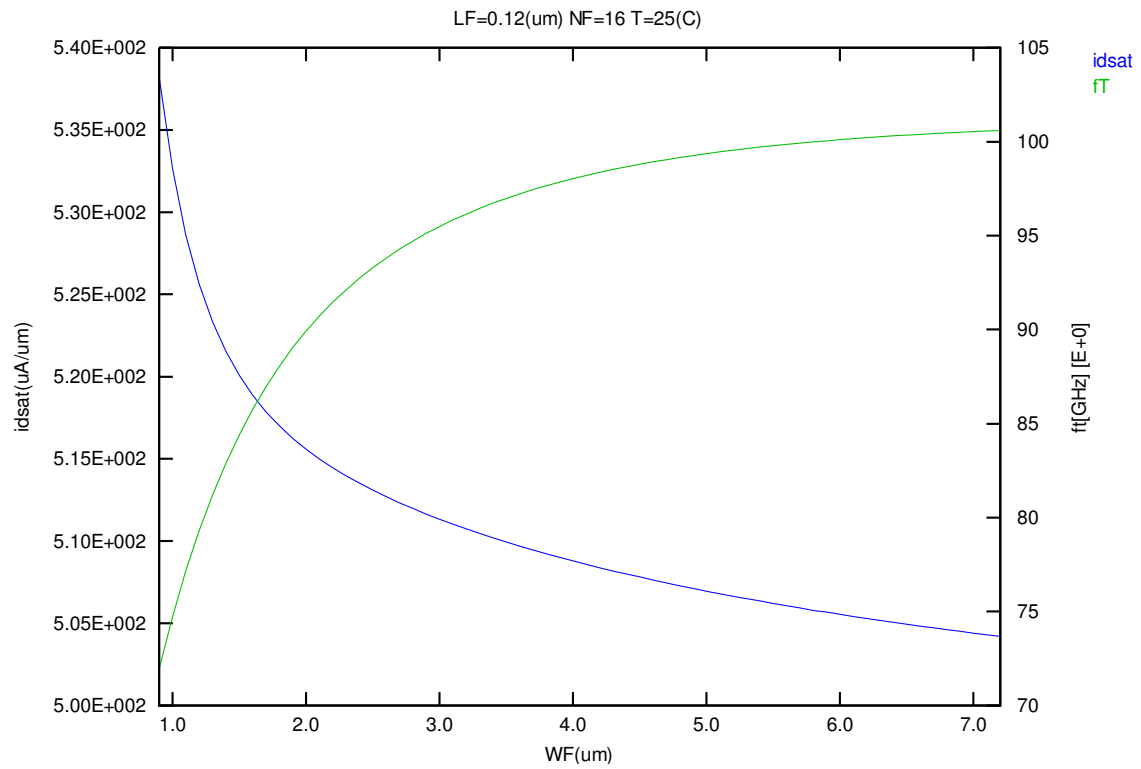


Figure 2-2 f_t and I_{dsat} vs. channel width

2.1.4 DC & RF Fitting curves for individual devices

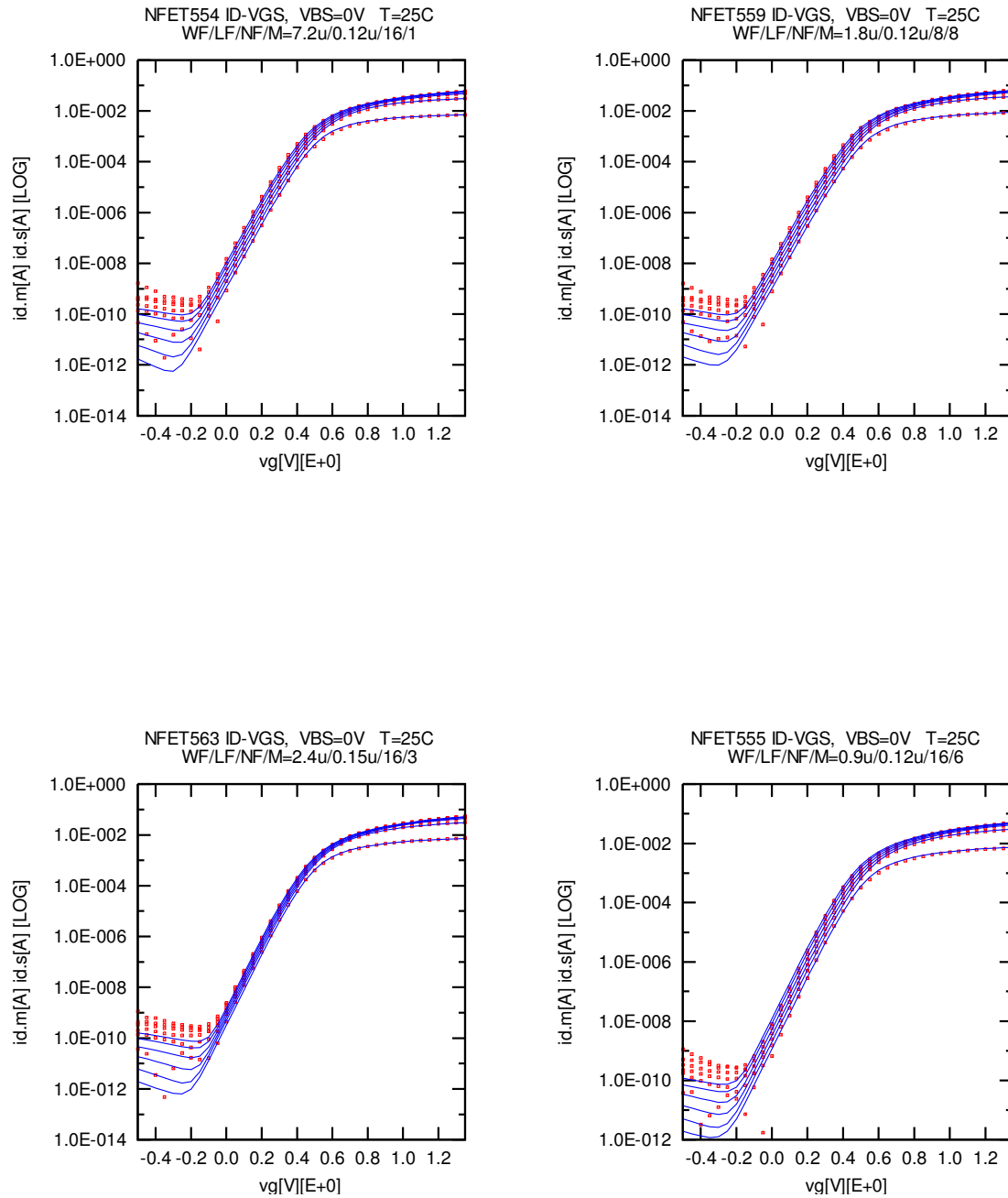


Figure 2-3 Id-Vg (25°C)

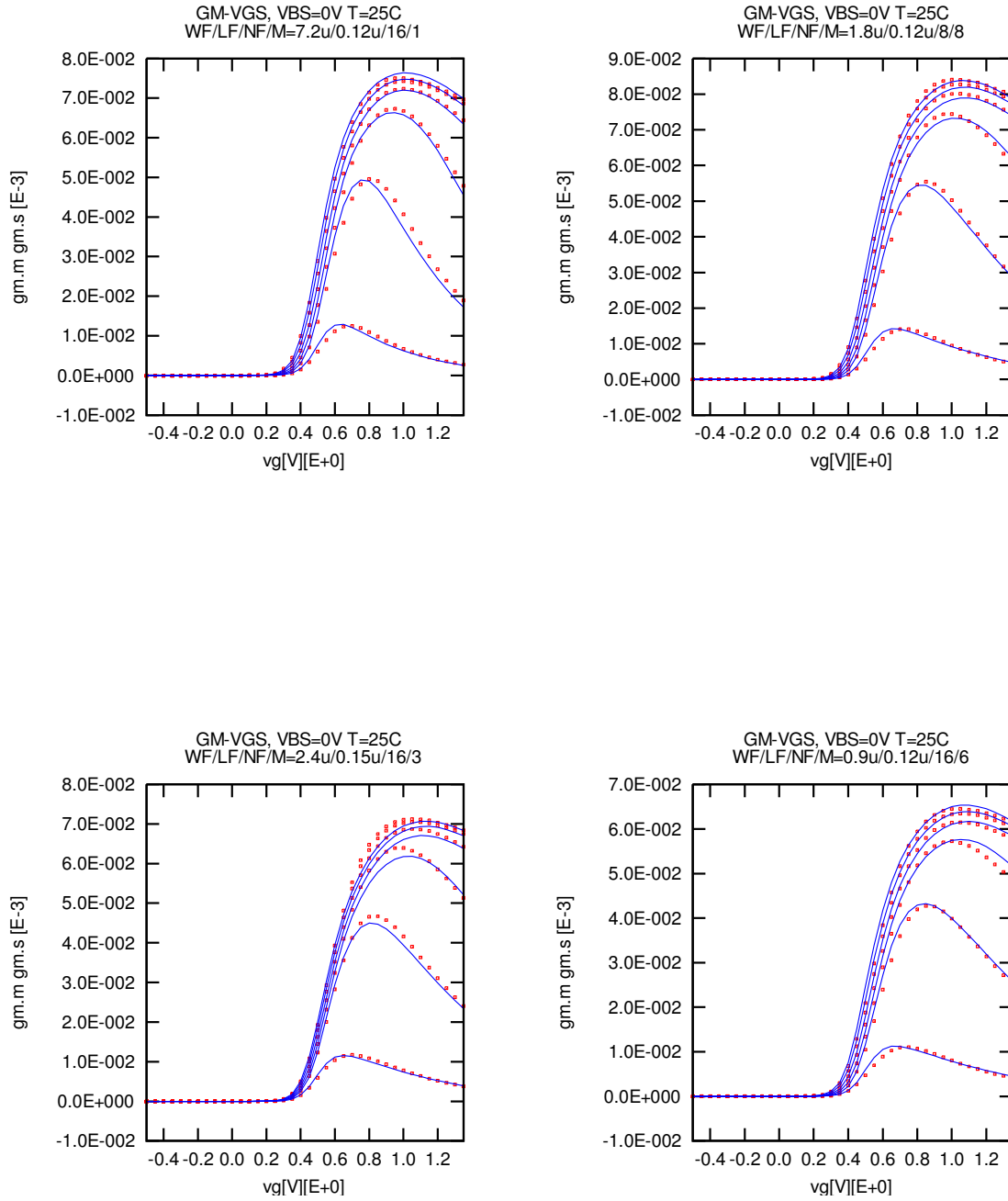


Figure 2-4 Gm-Vg (25°C)

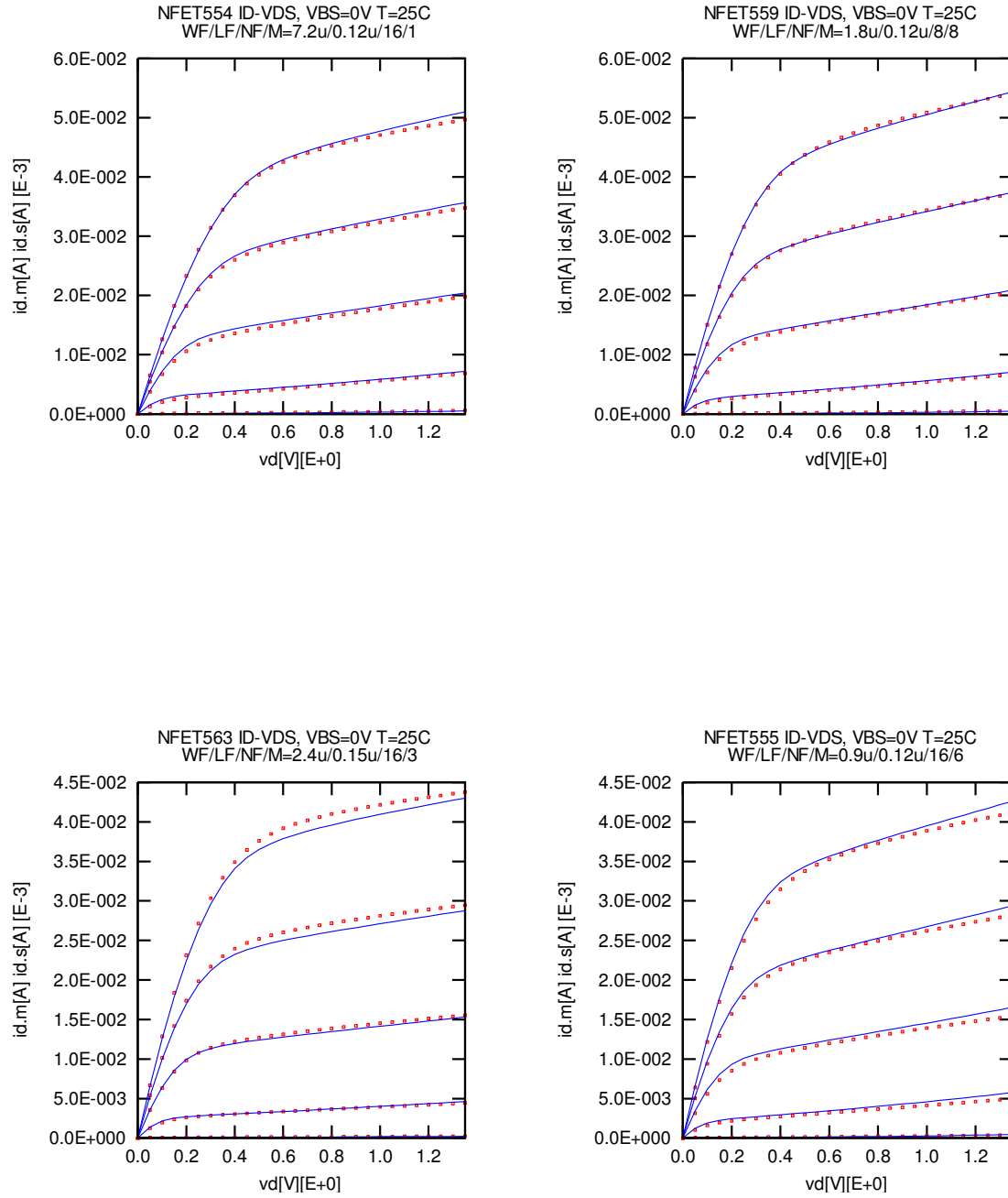


Figure 2-5 Id-Vd (25°C)

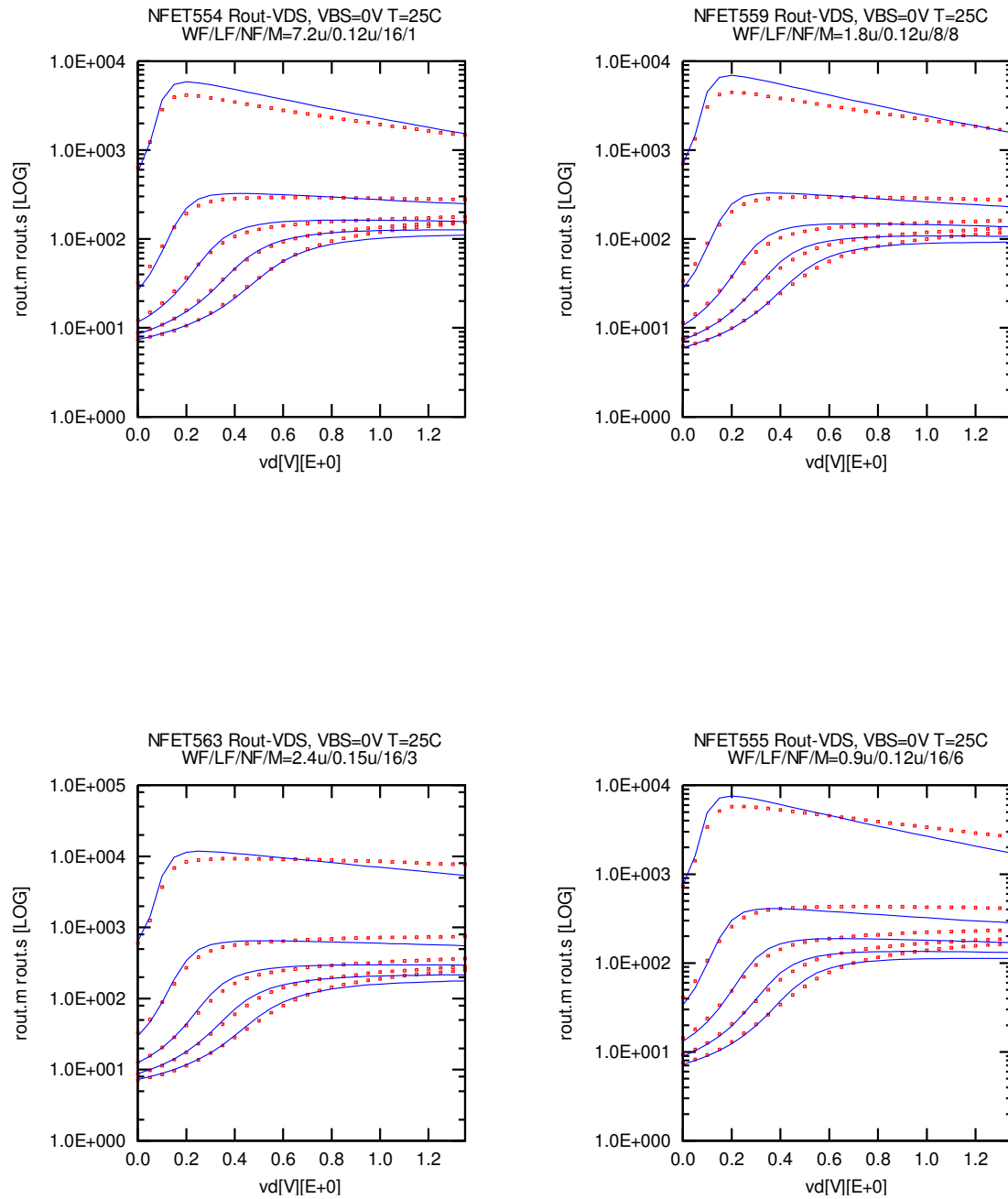


Figure 2-6 Rout-Vd (25°C)

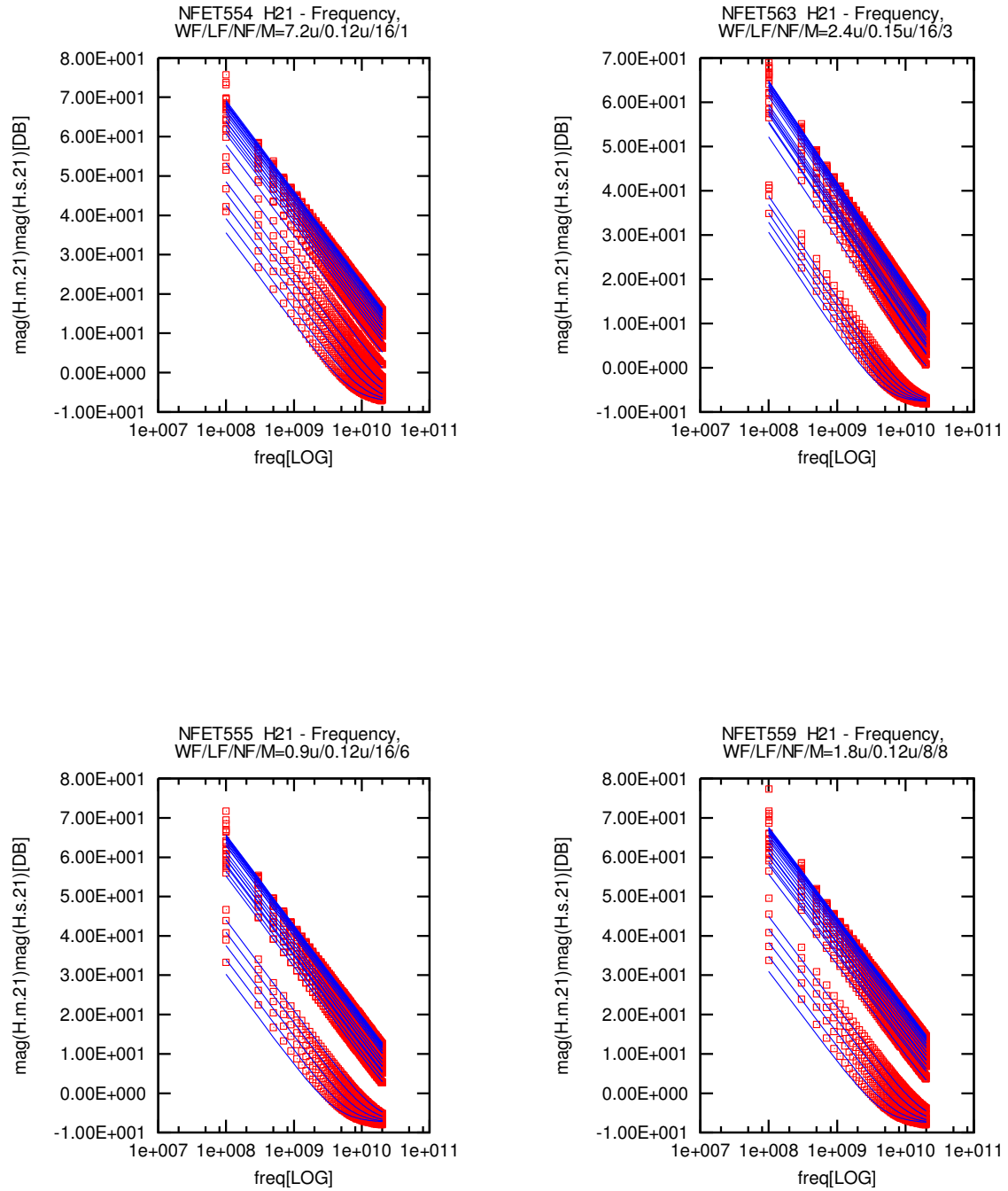


Figure 2-7 Mag(H21)-Freq (25°C)

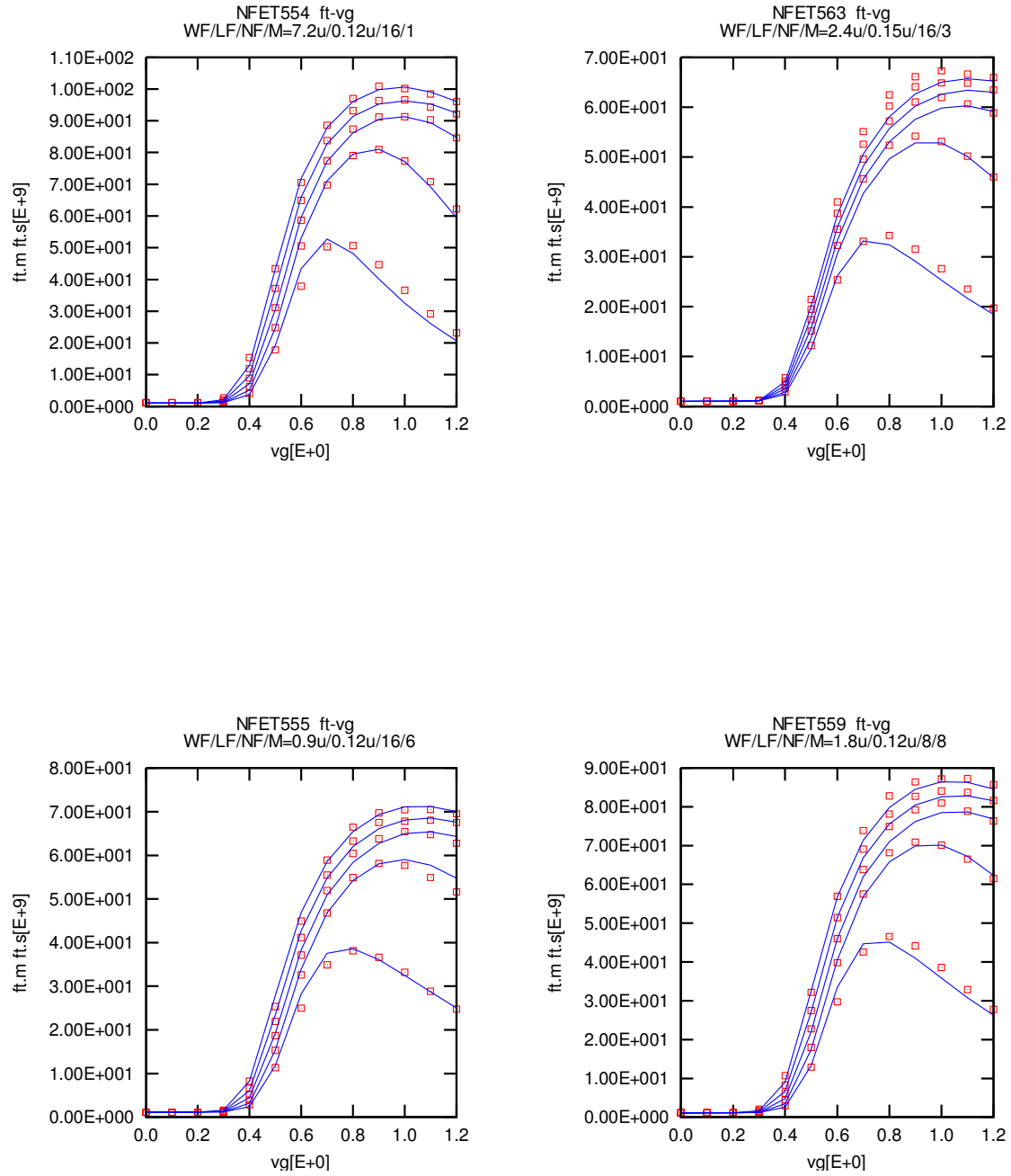


Figure 2-8 fT-Vg (25°C)

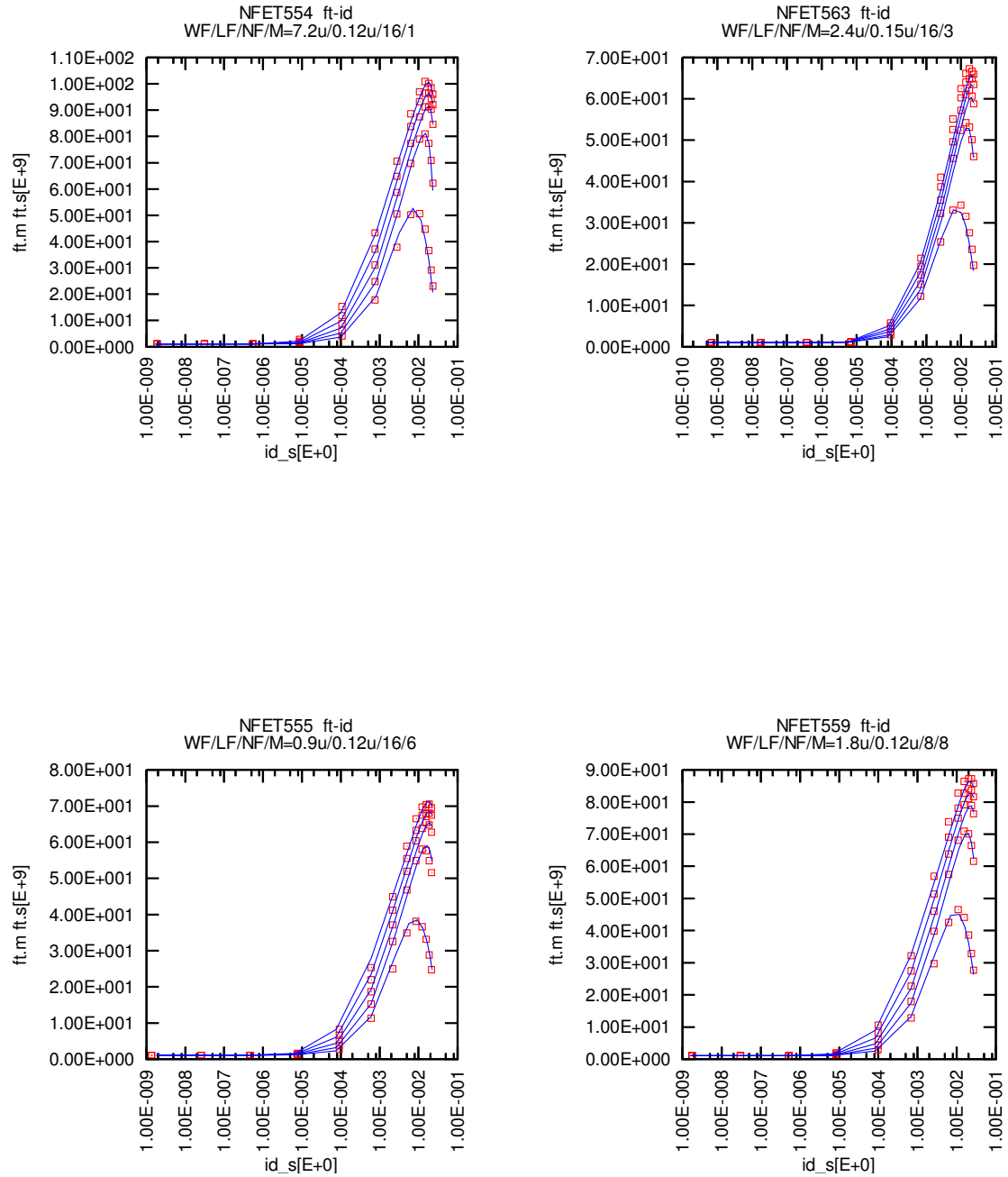
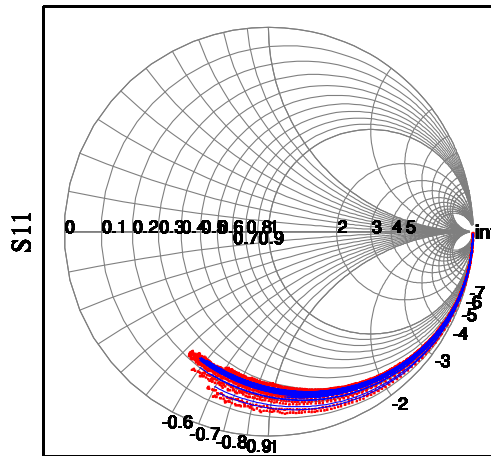
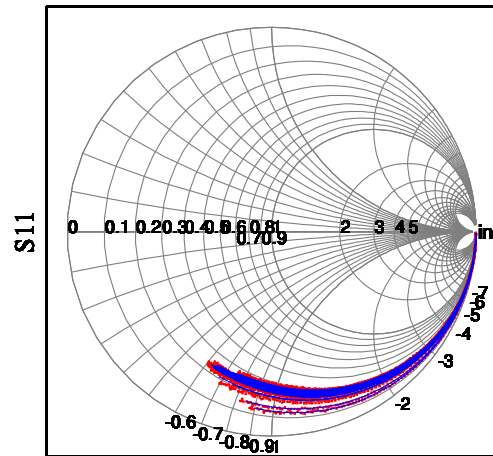


Figure 2-9 f_t - I_D (25°C)

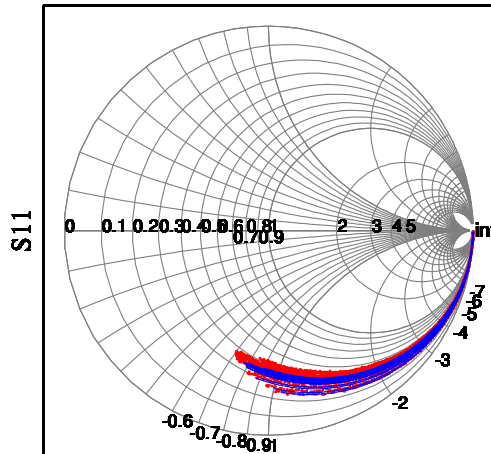
NFET563 S11 WF/LF/NF/M=2.4u/0.15u/16/3



NFET555 S11 WF/LF/NF/M=0.9u/0.12u/16/6



NFET554 S11 WF/LF/NF/M=7.2u/0.12u/16/1



NFET559 S11 WF/LF/NF/M=1.8u/0.12u/8/8

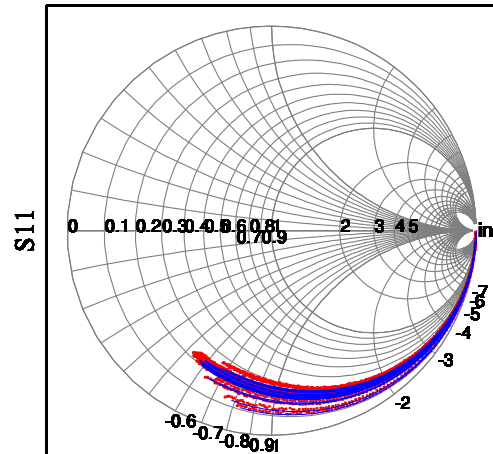


Figure 2-10 S11 (25°C)

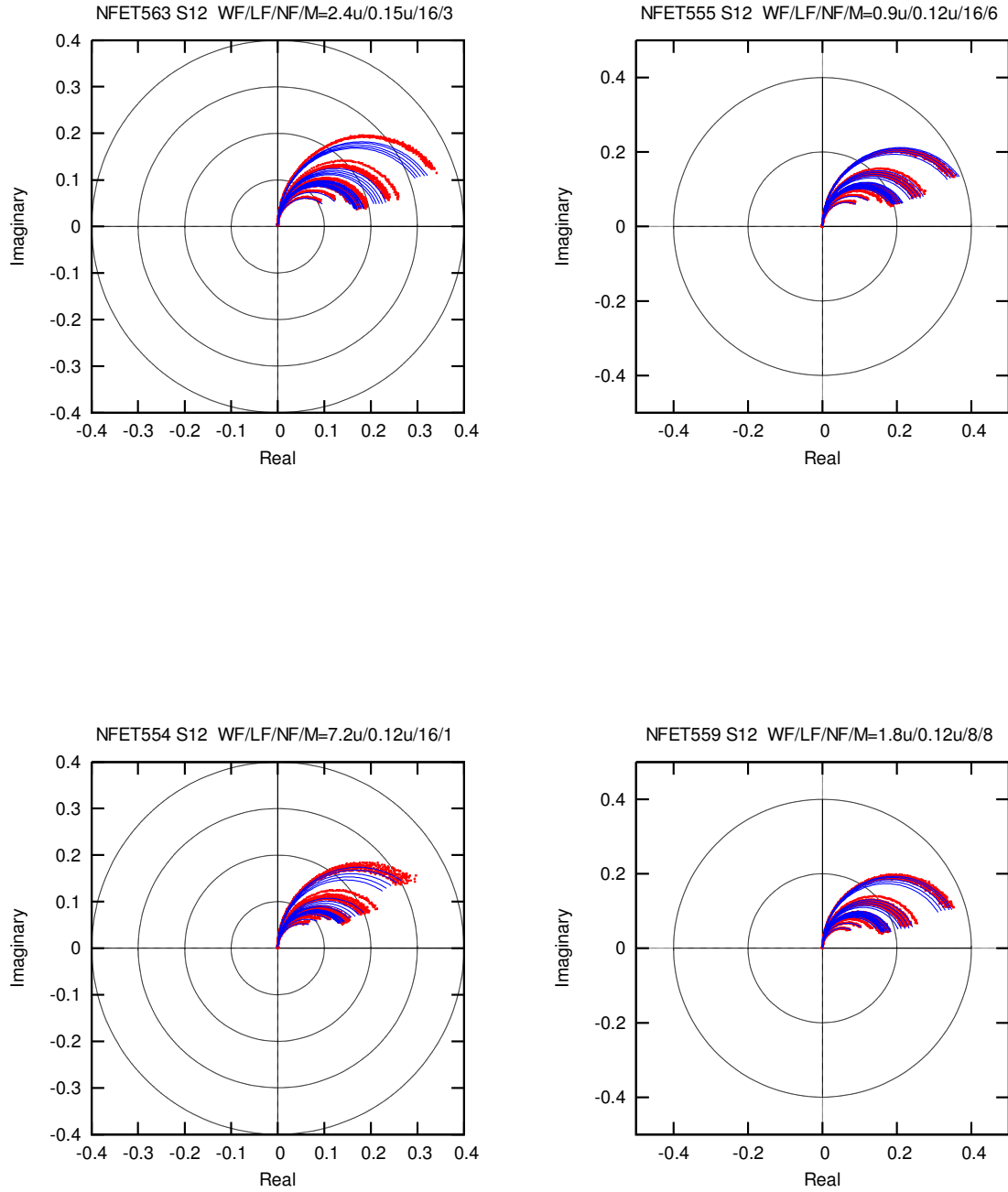


Figure 2-11 S12 (25°C)

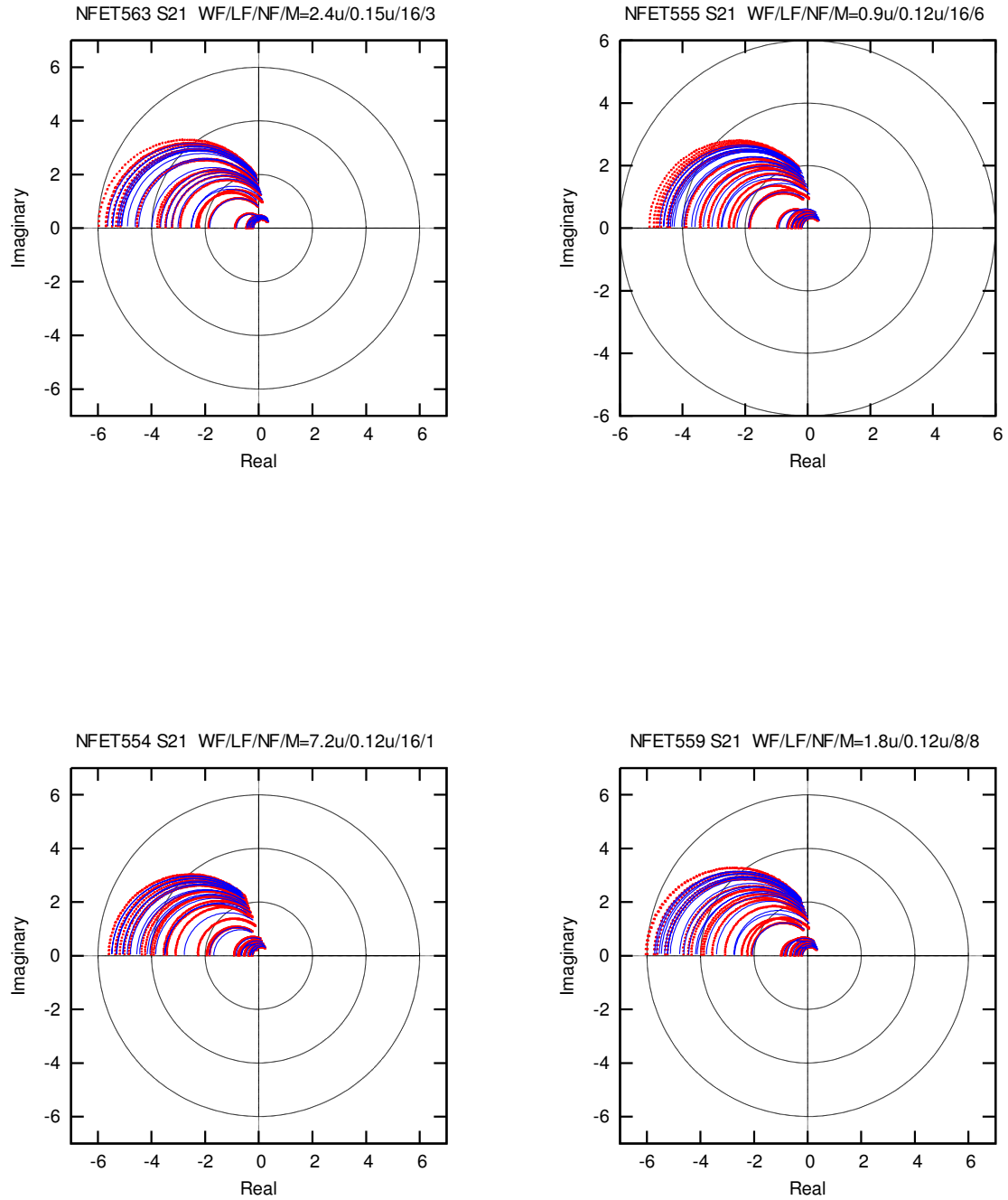
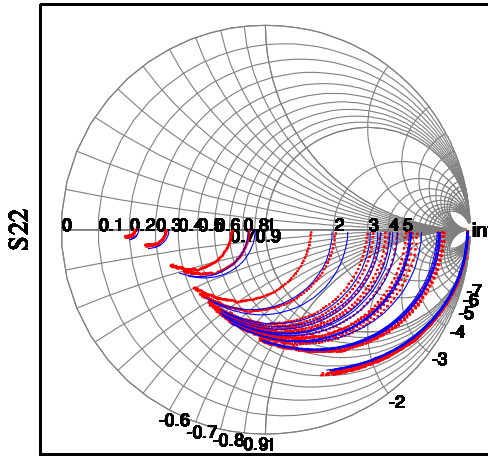
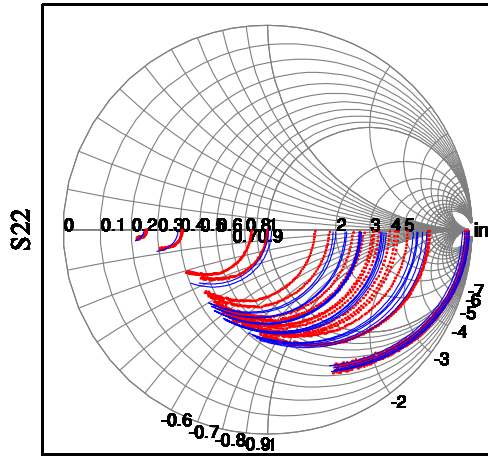


Figure 2-12 S21 (25°C)

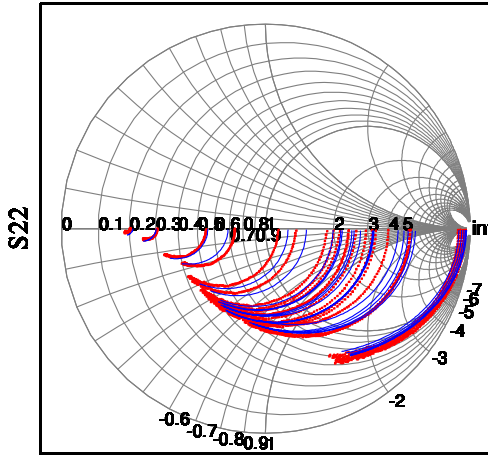
NFET563 S22 WF/LF/NF/M=2.4u/0.15u/16/3



NFET555 S22 WF/LF/NF/M=0.9u/0.12u/16/6



NFET554 S22 WF/LF/NF/M=7.2u/0.12u/16/1



NFET559 S22 WF/LF/NF/M=1.8u/0.12u/8/8

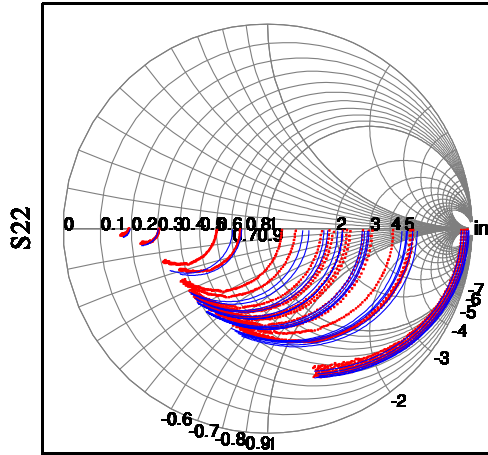


Figure 2-13 S22 (25°C)

2.2 PMOS Model

2.2.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max:Vd=-1.2V, sweep Vg
vth : Linear Vth extracted @ Vd=-0.05V @
Id=-70nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=-1.2V using fixed current
Id=-70nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ 1.2V
Idsat : Drain current extracted @ Vd=Vg=-1.2V
Idlin : Drain current extracted @ Vd=-0.05, Vg=-1.2V
Ioff : Drain current extracted @ Vg=0V, Vd=-1.2V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	16	16	22	16	16	16
SS	ft(GHz)	12.438	42.669	36.173	30.77	NA	NA	NA	NA
	idsat(uA/um)	-7.27E+01	-1.70E+02	-1.59E+02	-1.55E+02	-6.93E+01	-1.55E+02	-1.44E+02	-1.40E+02
	vtlin(V)	-4.64E-01	-4.65E-01	-4.82E-01	-4.92E-01	-4.66E-01	-4.85E-01	-5.00E-01	-5.07E-01
	vtsat(V)	-4.40E-01	-3.99E-01	-4.14E-01	-4.23E-01	-4.40E-01	-4.13E-01	-4.28E-01	-4.35E-01
	body_effect(V)	-2.68E-01	-1.66E-01	-1.65E-01	-1.61E-01	-2.67E-01	-2.17E-01	-2.11E-01	-2.08E-01
	ioff(pA/um)	-1.23E+00	-2.12E+01	-1.31E+01	-1.05E+01	-2.07E+00	-1.83E+01	-1.22E+01	-1.01E+01
SNFP	ft(GHz)	13.969	55.379	49.789	44.406	NA	NA	NA	NA
	idsat(uA/um)	-8.61E+01	-2.24E+02	-2.23E+02	-2.27E+02	-8.16E+01	-2.11E+02	-2.05E+02	-2.04E+02
	vtlin(V)	-3.89E-01	-3.71E-01	-3.74E-01	-3.73E-01	-3.91E-01	-3.91E-01	-3.91E-01	-3.89E-01
	vtsat(V)	-3.65E-01	-3.01E-01	-3.01E-01	-3.00E-01	-3.66E-01	-3.11E-01	-3.12E-01	-3.12E-01
	body_effect(V)	-2.59E-01	-7.92E-02	-8.89E-02	-9.19E-02	-2.61E-01	-1.97E-01	-1.94E-01	-1.92E-01
	ioff(pA/um)	-9.61E+00	-4.48E+02	-3.74E+02	-3.56E+02	-1.51E+01	-2.74E+02	-2.51E+02	-2.44E+02
TT	ft(GHz)	13.639	51.92	45.283	39.414	NA	NA	NA	NA
	idsat(uA/um)	-8.19E+01	-2.04E+02	-2.01E+02	-2.02E+02	-7.73E+01	-1.92E+02	-1.85E+02	-1.83E+02
	vtlin(V)	-4.14E-01	-4.02E-01	-4.09E-01	-4.12E-01	-4.16E-01	-4.22E-01	-4.27E-01	-4.28E-01
	vtsat(V)	-3.90E-01	-3.31E-01	-3.37E-01	-3.39E-01	-3.91E-01	-3.44E-01	-3.50E-01	-3.52E-01
	body_effect(V)	-2.60E-01	-1.30E-01	-1.33E-01	-1.32E-01	-2.61E-01	-2.01E-01	-1.97E-01	-1.95E-01
	ioff(pA/um)	-4.52E+00	-1.50E+02	-1.12E+02	-9.86E+01	-7.49E+00	-1.12E+02	-9.07E+01	-8.22E+01
FNFP	ft(GHz)	13.125	46.831	40.602	35.414	NA	NA	NA	NA
	idsat(uA/um)	-7.76E+01	-1.83E+02	-1.77E+02	-1.78E+02	-7.33E+01	-1.74E+02	-1.65E+02	-1.62E+02
	vtlin(V)	-4.39E-01	-4.34E-01	-4.45E-01	-4.52E-01	-4.41E-01	-4.54E-01	-4.63E-01	-4.67E-01
	vtsat(V)	-4.15E-01	-3.64E-01	-3.75E-01	-3.81E-01	-4.15E-01	-3.79E-01	-3.88E-01	-3.93E-01
	body_effect(V)	-2.60E-01	-1.66E-01	-1.64E-01	-1.61E-01	-2.61E-01	-2.07E-01	-2.02E-01	-2.00E-01
	ioff(pA/um)	-2.23E+00	-4.88E+01	-3.31E+01	-2.73E+01	-3.78E+00	-4.41E+01	-3.23E+01	-2.78E+01
FF	ft(GHz)	14.608	61.32	54.375	47.732	NA	NA	NA	NA
	idsat(uA/um)	-9.15E+01	-2.41E+02	-2.45E+02	-2.50E+02	-8.64E+01	-2.29E+02	-2.25E+02	-2.26E+02
	vtlin(V)	-3.64E-01	-3.40E-01	-3.37E-01	-3.34E-01	-3.65E-01	-3.59E-01	-3.54E-01	-3.50E-01
	vtsat(V)	-3.40E-01	-2.68E-01	-2.64E-01	-2.60E-01	-3.41E-01	-2.80E-01	-2.76E-01	-2.73E-01
	body_effect(V)	-2.51E-01	-8.33E-02	-9.14E-02	-9.30E-02	-2.56E-01	-1.92E-01	-1.89E-01	-1.88E-01
	ioff(pA/um)	-1.97E+01	-9.26E+02	-9.04E+02	-9.43E+02	-3.00E+01	-5.98E+02	-6.28E+02	-6.59E+02

Table 2-5 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	16	16	22	16	16	16
SS	ft(GHz)	11.045	35.787	30.067	25.331	NA	NA	NA	NA
	idsat(uA/um)	-6.69E+01	-1.52E+02	-1.42E+02	-1.39E+02	-6.78E+01	-1.52E+02	-1.41E+02	-1.37E+02
	vtlin(V)	-4.01E-01	-3.98E-01	-4.18E-01	-4.30E-01	-4.02E-01	-4.14E-01	-4.32E-01	-4.42E-01
	vt sat(V)	-3.69E-01	-3.22E-01	-3.40E-01	-3.50E-01	-3.69E-01	-3.33E-01	-3.50E-01	-3.59E-01
	body_effect(V)	-2.77E-01	-1.79E-01	-1.76E-01	-1.73E-01	-2.76E-01	-2.27E-01	-2.20E-01	-2.17E-01
	ioff(pA/um)	-1.14E+02	-1.35E+03	-8.69E+02	-6.77E+02	-1.73E+02	-1.24E+03	-8.34E+02	-6.72E+02
SNFP	ft(GHz)	12.234	44.603	39.23	34.317	NA	NA	NA	NA
	idsat(uA/um)	-7.74E+01	-1.94E+02	-1.91E+02	-1.92E+02	-7.67E+01	-1.98E+02	-1.92E+02	-1.90E+02
	vtlin(V)	-3.25E-01	-3.02E-01	-3.07E-01	-3.08E-01	-3.24E-01	-3.15E-01	-3.18E-01	-3.19E-01
	vt sat(V)	-2.93E-01	-2.21E-01	-2.23E-01	-2.23E-01	-2.92E-01	-2.26E-01	-2.30E-01	-2.30E-01
	body_effect(V)	-2.69E-01	-9.31E-02	-1.03E-01	-1.06E-01	-2.71E-01	-2.08E-01	-2.05E-01	-2.03E-01
	ioff(pA/um)	-6.41E+02	-1.26E+04	-1.10E+04	-1.05E+04	-8.72E+02	-1.01E+04	-9.10E+03	-8.76E+03
TT	ft(GHz)	12.069	42.68	36.611	31.403	NA	NA	NA	NA
	idsat(uA/um)	-7.44E+01	-1.79E+02	-1.75E+02	-1.75E+02	-7.36E+01	-1.83E+02	-1.75E+02	-1.73E+02
	vtlin(V)	-3.51E-01	-3.34E-01	-3.43E-01	-3.48E-01	-3.50E-01	-3.48E-01	-3.56E-01	-3.59E-01
	vt sat(V)	-3.19E-01	-2.53E-01	-2.61E-01	-2.64E-01	-3.18E-01	-2.61E-01	-2.69E-01	-2.72E-01
	body_effect(V)	-2.70E-01	-1.44E-01	-1.46E-01	-1.45E-01	-2.71E-01	-2.11E-01	-2.08E-01	-2.06E-01
	ioff(pA/um)	-3.54E+02	-5.81E+03	-4.61E+03	-4.13E+03	-5.04E+02	-5.10E+03	-4.18E+03	-3.80E+03
FNFP	ft(GHz)	11.733	39.226	33.624	29.022	NA	NA	NA	NA
	idsat(uA/um)	-7.13E+01	-1.63E+02	-1.57E+02	-1.57E+02	-7.09E+01	-1.68E+02	-1.59E+02	-1.56E+02
	vtlin(V)	-3.76E-01	-3.66E-01	-3.80E-01	-3.88E-01	-3.76E-01	-3.81E-01	-3.93E-01	-3.99E-01
	vt sat(V)	-3.44E-01	-2.87E-01	-3.00E-01	-3.06E-01	-3.43E-01	-2.97E-01	-3.09E-01	-3.15E-01
	body_effect(V)	-2.70E-01	-1.78E-01	-1.76E-01	-1.73E-01	-2.71E-01	-2.17E-01	-2.12E-01	-2.09E-01
	ioff(pA/um)	-1.95E+02	-2.58E+03	-1.85E+03	-1.55E+03	-2.91E+02	-2.49E+03	-1.87E+03	-1.61E+03
FF	ft(GHz)	12.888	49.395	42.648	36.573	NA	NA	NA	NA
	idsat(uA/um)	-8.21E+01	-2.08E+02	-2.08E+02	-2.10E+02	-8.05E+01	-2.14E+02	-2.09E+02	-2.09E+02
	vtlin(V)	-3.00E-01	-2.70E-01	-2.70E-01	-2.68E-01	-2.98E-01	-2.83E-01	-2.81E-01	-2.78E-01
	vt sat(V)	-2.68E-01	-1.88E-01	-1.85E-01	-1.82E-01	-2.66E-01	-1.94E-01	-1.92E-01	-1.90E-01
	body_effect(V)	-2.62E-01	-9.73E-02	-1.05E-01	-1.07E-01	-2.66E-01	-2.03E-01	-2.01E-01	-1.99E-01
	ioff(pA/um)	-1.10E+03	-2.20E+04	-2.16E+04	-2.21E+04	-1.48E+03	-1.83E+04	-1.84E+04	-1.87E+04

Table 2-6 RF & Logic Corner Table at 125 °C

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	0.24	0.12	0.12	0.12	0.24	0.12	0.12	0.12
	WF (um)	9.6	9.6	2.4	1.6	9.6	9.6	2.4	1.6
	NF	22	16	16	16	22	16	16	16
SS	ft(GHz)	13.685	48.944	41.833	35.908	NA	NA	NA	NA
	idsat(uA/um)	-7.85E+01	-1.86E+02	-1.74E+02	-1.71E+02	-7.17E+01	-1.60E+02	-1.49E+02	-1.45E+02
	vtlin(V)	-5.18E-01	-5.21E-01	-5.36E-01	-5.44E-01	-5.20E-01	-5.44E-01	-5.56E-01	-5.62E-01
	vtsat(V)	-4.98E-01	-4.59E-01	-4.73E-01	-4.81E-01	-4.98E-01	-4.75E-01	-4.89E-01	-4.95E-01
	body_effect(V)	-2.63E-01	-1.57E-01	-1.56E-01	-1.53E-01	-2.62E-01	-2.13E-01	-2.06E-01	-2.03E-01
	ioff(pA/um)	-3.12E-01	-3.91E-01	-1.24E+00	-1.83E+00	-2.82E-01	-3.43E-01	-1.15E+00	-1.69E+00
SNFP	ft(GHz)	15.502	65.665	60.304	54.842	NA	NA	NA	NA
	idsat(uA/um)	-9.51E+01	-2.54E+02	-2.56E+02	-2.63E+02	-8.72E+01	-2.24E+02	-2.19E+02	-2.19E+02
	vtlin(V)	-4.43E-01	-4.27E-01	-4.27E-01	-4.26E-01	-4.46E-01	-4.51E-01	-4.49E-01	-4.46E-01
	vtsat(V)	-4.23E-01	-3.61E-01	-3.61E-01	-3.58E-01	-4.25E-01	-3.76E-01	-3.75E-01	-3.74E-01
	body_effect(V)	-2.54E-01	-6.76E-02	-7.79E-02	-8.17E-02	-2.56E-01	-1.92E-01	-1.89E-01	-1.87E-01
	ioff(pA/um)	-3.93E-01	-5.68E+00	-5.43E+00	-5.76E+00	-3.45E-01	-2.57E+00	-3.23E+00	-3.75E+00
TT	ft(GHz)	15.011	60.488	53.589	47.316	NA	NA	NA	NA
	idsat(uA/um)	-8.95E+01	-2.29E+02	-2.26E+02	-2.30E+02	-8.17E+01	-2.02E+02	-1.95E+02	-1.94E+02
	vtlin(V)	-4.68E-01	-4.58E-01	-4.63E-01	-4.65E-01	-4.71E-01	-4.82E-01	-4.84E-01	-4.85E-01
	vtsat(V)	-4.48E-01	-3.91E-01	-3.96E-01	-3.98E-01	-4.49E-01	-4.09E-01	-4.12E-01	-4.13E-01
	body_effect(V)	-2.54E-01	-1.20E-01	-1.23E-01	-1.23E-01	-2.56E-01	-1.96E-01	-1.92E-01	-1.90E-01
	ioff(pA/um)	-3.71E-01	-1.50E+00	-2.04E+00	-2.52E+00	-3.04E-01	-9.81E-01	-1.66E+00	-2.15E+00
FNFP	ft(GHz)	14.329	53.663	47.019	41.424	NA	NA	NA	NA
	idsat(uA/um)	-8.39E+01	-2.02E+02	-1.97E+02	-1.98E+02	-7.65E+01	-1.80E+02	-1.72E+02	-1.69E+02
	vtlin(V)	-4.93E-01	-4.90E-01	-5.00E-01	-5.05E-01	-4.95E-01	-5.13E-01	-5.20E-01	-5.23E-01
	vtsat(V)	-4.73E-01	-4.24E-01	-4.34E-01	-4.39E-01	-4.74E-01	-4.42E-01	-4.50E-01	-4.54E-01
	body_effect(V)	-2.55E-01	-1.57E-01	-1.55E-01	-1.52E-01	-2.56E-01	-2.02E-01	-1.97E-01	-1.95E-01
	ioff(pA/um)	-3.64E-01	-5.91E-01	-1.39E+00	-1.96E+00	-2.88E-01	-4.84E-01	-1.25E+00	-1.78E+00
FF	ft(GHz)	16.073	72.594	66.081	59.4	NA	NA	NA	NA
	idsat(uA/um)	-1.01E+02	-2.75E+02	-2.84E+02	-2.94E+02	-9.29E+01	-2.45E+02	-2.43E+02	-2.44E+02
	vtlin(V)	-4.17E-01	-3.95E-01	-3.91E-01	-3.87E-01	-4.21E-01	-4.20E-01	-4.13E-01	-4.08E-01
	vtsat(V)	-3.98E-01	-3.29E-01	-3.24E-01	-3.19E-01	-4.00E-01	-3.46E-01	-3.40E-01	-3.36E-01
	body_effect(V)	-2.46E-01	-7.22E-02	-8.08E-02	-8.29E-02	-2.50E-01	-1.87E-01	-1.83E-01	-1.82E-01
	ioff(pA/um)	-6.22E-01	-1.42E+01	-1.47E+01	-1.62E+01	-4.45E-01	-6.64E+00	-8.20E+00	-9.38E+00

Table 2-7 RF & Logic Corner Table at -50 °C

2.2.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
PFET601	0.12	1.6	16	6	1	-0.417	-0.413	-0.33	-0.339	-196.14	-199.91	-126.1	-98.59	38.505	38.449
PFET602	0.12	2.4	16	4	1	-0.418	-0.41	-0.33	-0.337	-196.46	-197.44	-127.82	-112.45	42.449	44.054
PFET603	0.12	4.8	16	2	1	-0.417	-0.405	-0.329	-0.334	-198.21	-195.51	-133.55	-133.31	47.322	47.756
PFET604	0.12	9.6	16	1	1	-0.414	-0.402	-0.325	-0.331	-195.5	-196.18	-154.09	-149.76	48.876	48.985
PFET605	0.15	2.4	16	4	1	-0.432	-0.427	-0.386	-0.38	-136.05	-132.39	-14.49	-15.669	26.963	27.207
PFET606	0.18	2.4	16	4	1	-0.424	-0.423	-0.39	-0.387	-110.11	-105.27	-9.97	-9.031	19.472	19.277
PFET607	0.24	2.4	16	4	1	-0.413	-0.415	-0.385	-0.389	-80.94	-77.38	-8.36	-5.704	11.767	11.61
PFET608	0.36	2.4	16	4	1	-0.403	-0.405	-0.38	-0.386	-53.56	-52.267	-6.15	-3.939	5.791	5.843
PFET609	0.12	2.4	2	18	1	-0.427	-0.41	-0.35	-0.337	-195.86	-198.79	-63.61	-112.94	38.679	34.673
PFET611	0.12	2.4	5	13	1	-0.422	-0.41	-0.34	-0.337	-194.52	-198.61	-93.43	-112.6	39.844	43.102
PFET612	0.12	2.4	8	8	1	-0.419	-0.41	-0.333	-0.337	-197.48	-197.89	-117.48	-112.52	41.197	43.938
PFET613	0.12	2.4	11	6	1	-0.419	-0.41	-0.333	-0.337	-196.39	-197.6	-121.63	-112.48	41.346	44.171
PFET619	0.12	4.8	4	8	1	-0.419	-0.405	-0.334	-0.334	-201.42	-197.27	-113.48	-133.42	45.732	47.037
PFET620	0.12	4.8	11	2	1	-0.417	-0.405	-0.329	-0.334	-199.64	-196.6	-140.09	-133.33	46.888	48.428
PFET621	0.12	4.8	22	1	1	-0.417	-0.405	-0.33	-0.334	-197.52	-195.88	-131.63	-133.31	46.415	47.954
PFET622	0.15	9.6	4	4	1	-0.432	-0.427	-0.388	-0.381	-141.83	-134.78	-14	-14.697	31.029	30.466
PFET623	0.18	9.6	11	1	1	-0.424	-0.423	-0.39	-0.389	-112.7	-107.85	-10.16	-7.743	22.445	22.119
PFET624	0.24	9.6	22	1	1	-0.417	-0.414	-0.389	-0.39	-81.26	-80.299	-7	-4.515	13.139	13.335
PFET625	0.12	2.4	4	1	1	-0.423	-0.41	-0.341	-0.337	-198.56	-199.29	-101.27	-112.66	34.963	42.3

Table 2-8 Summary Table for RF PMOS

Measure: Measurement result

Model: Simulation result

2.2.3 Simulation results for WF, LF trends

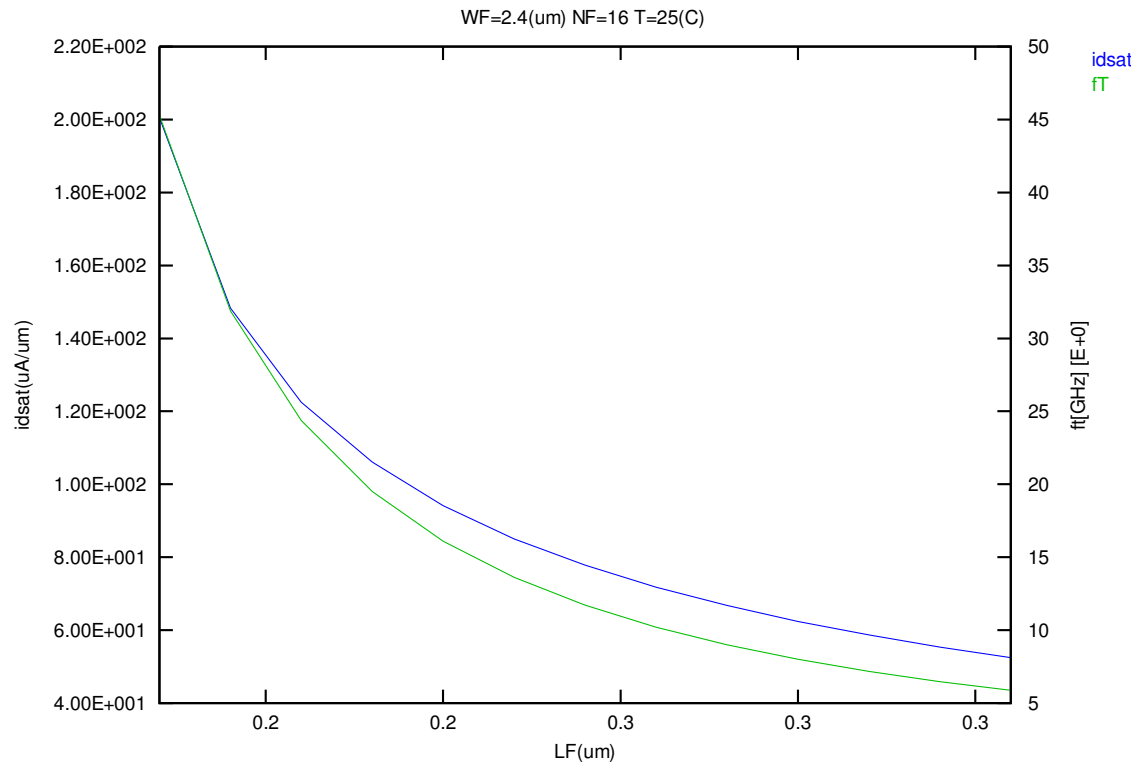


Figure 2-14 fT and Idsat vs. channel length

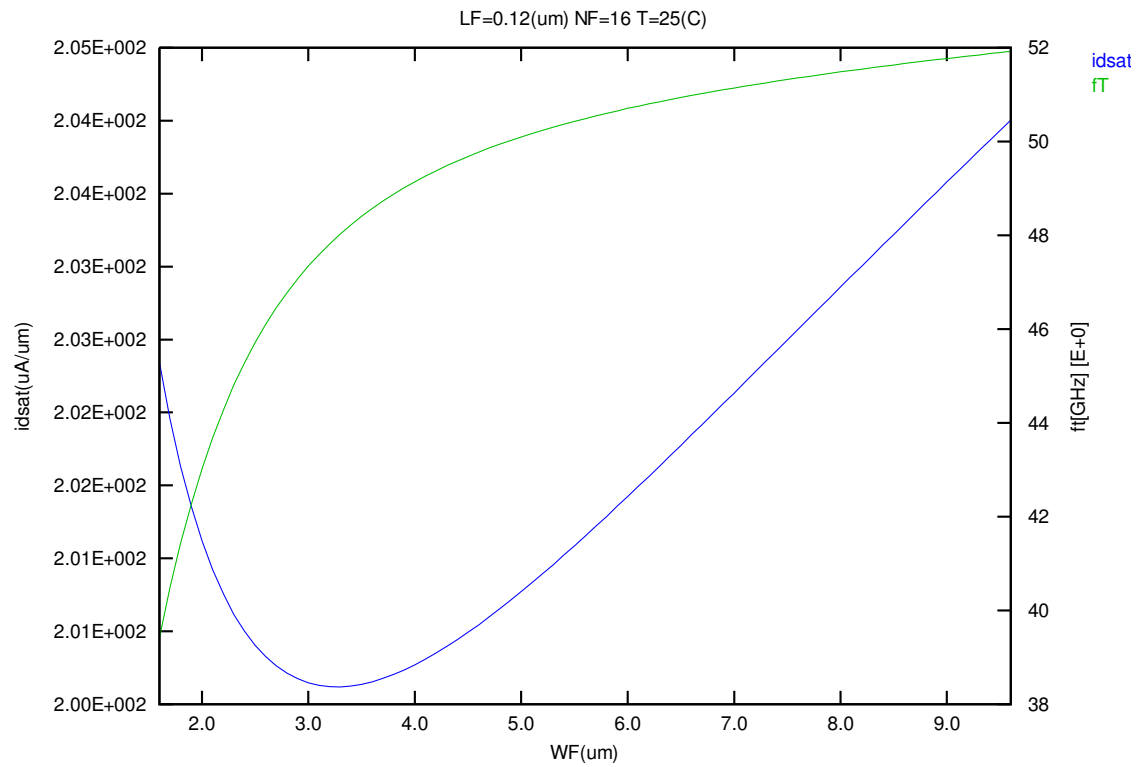


Figure 2-15 ft and Idsat vs. channel width

2.2.4 DC & RF Fitting curves for individual devices

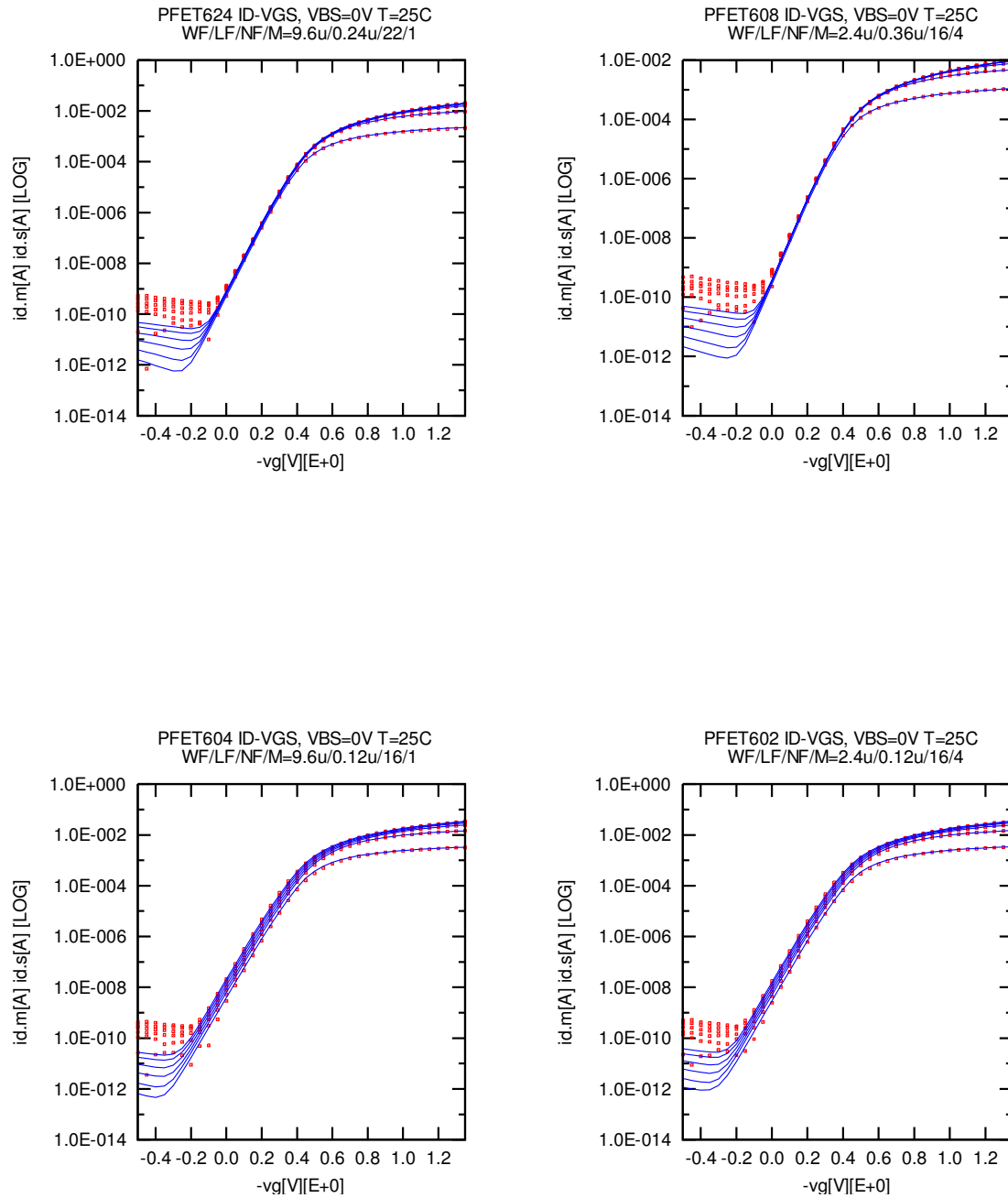


Figure 2-16 Id-Vg (25°C)

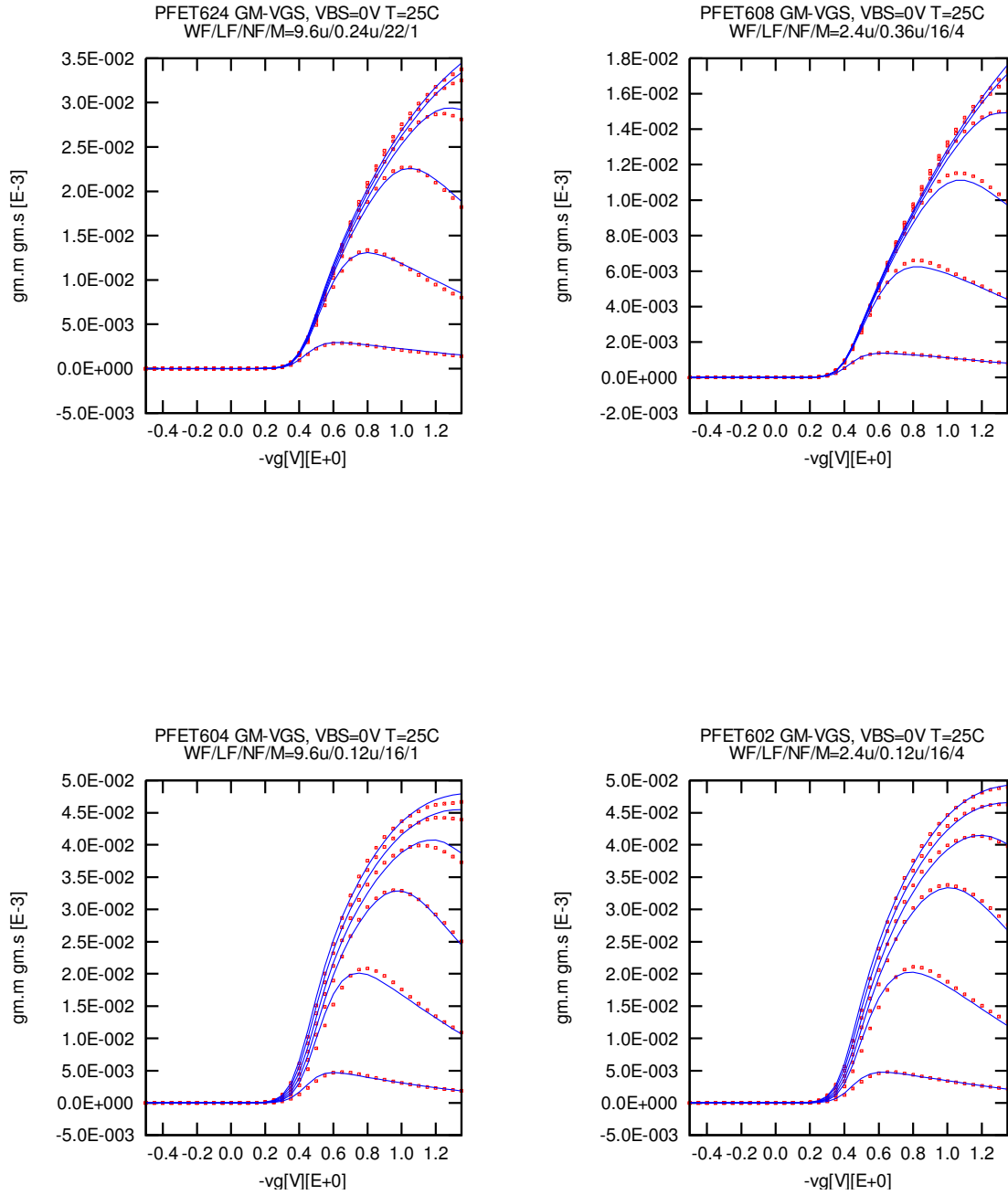


Figure 2-17 Gm-Vg (25°C)

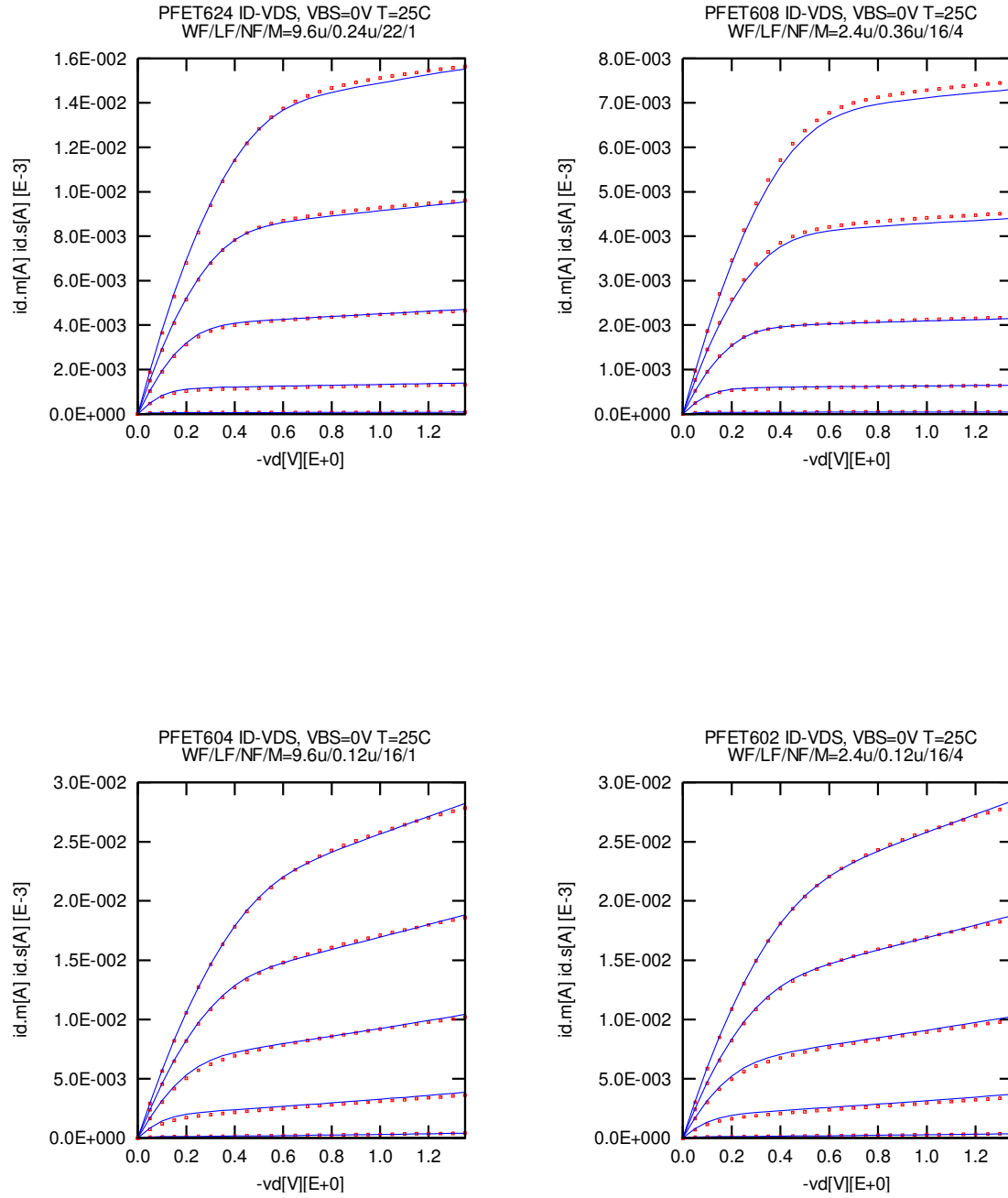


Figure 2-18 Id-Vd (25°C)

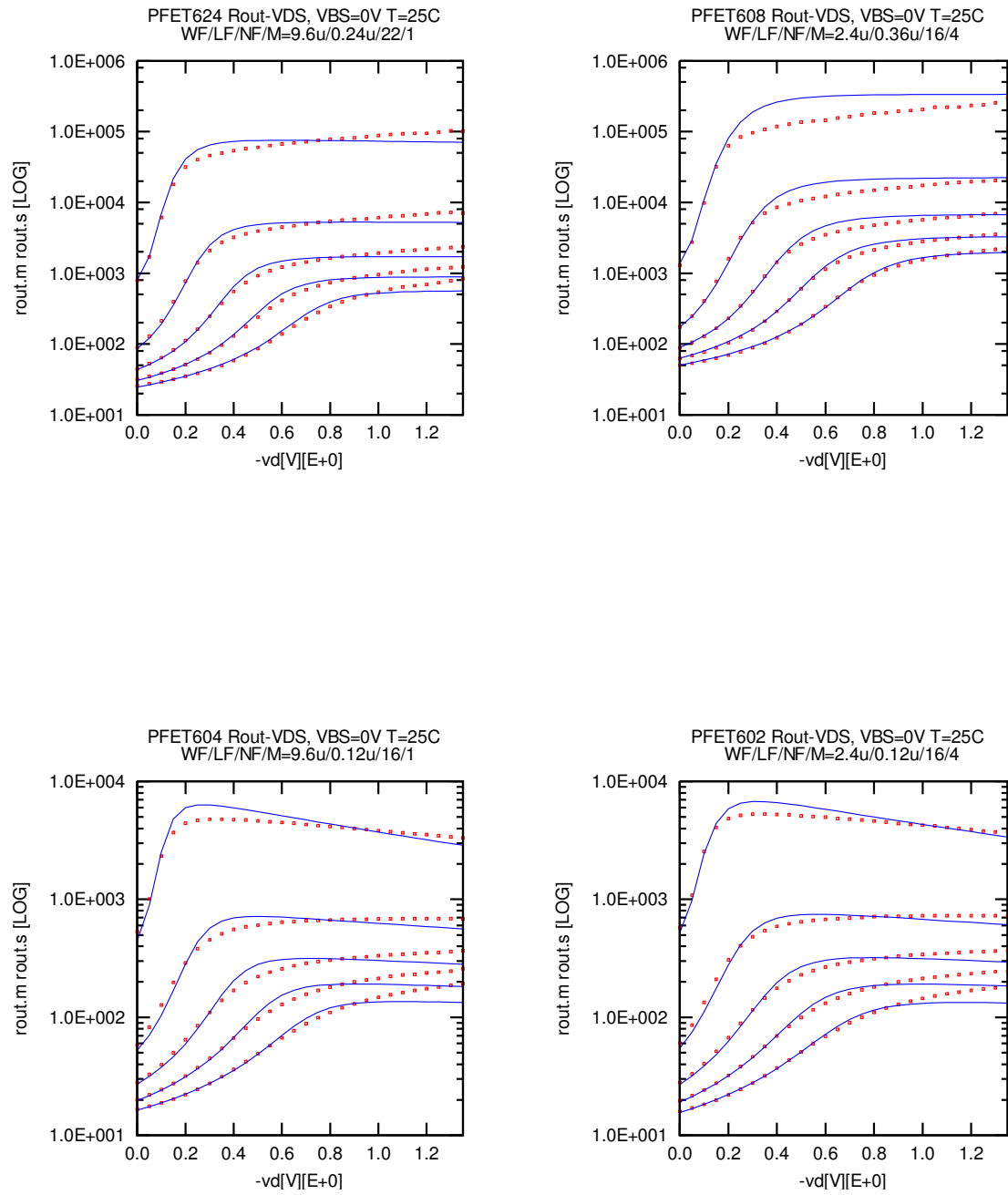


Figure 2-19 Rout-Vd (25°C)

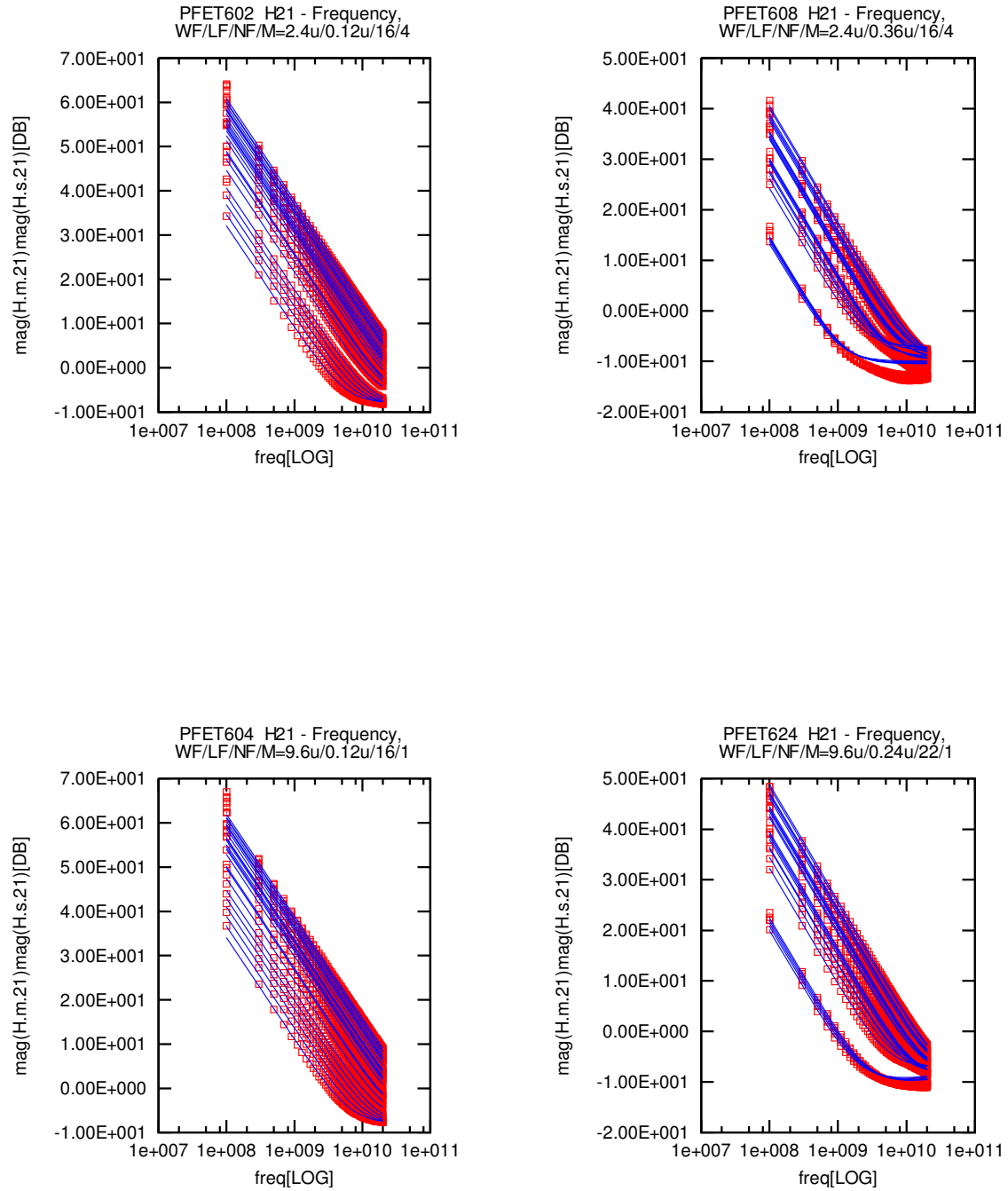


Figure 2-20 Mag(H21)-Freq (25°C)

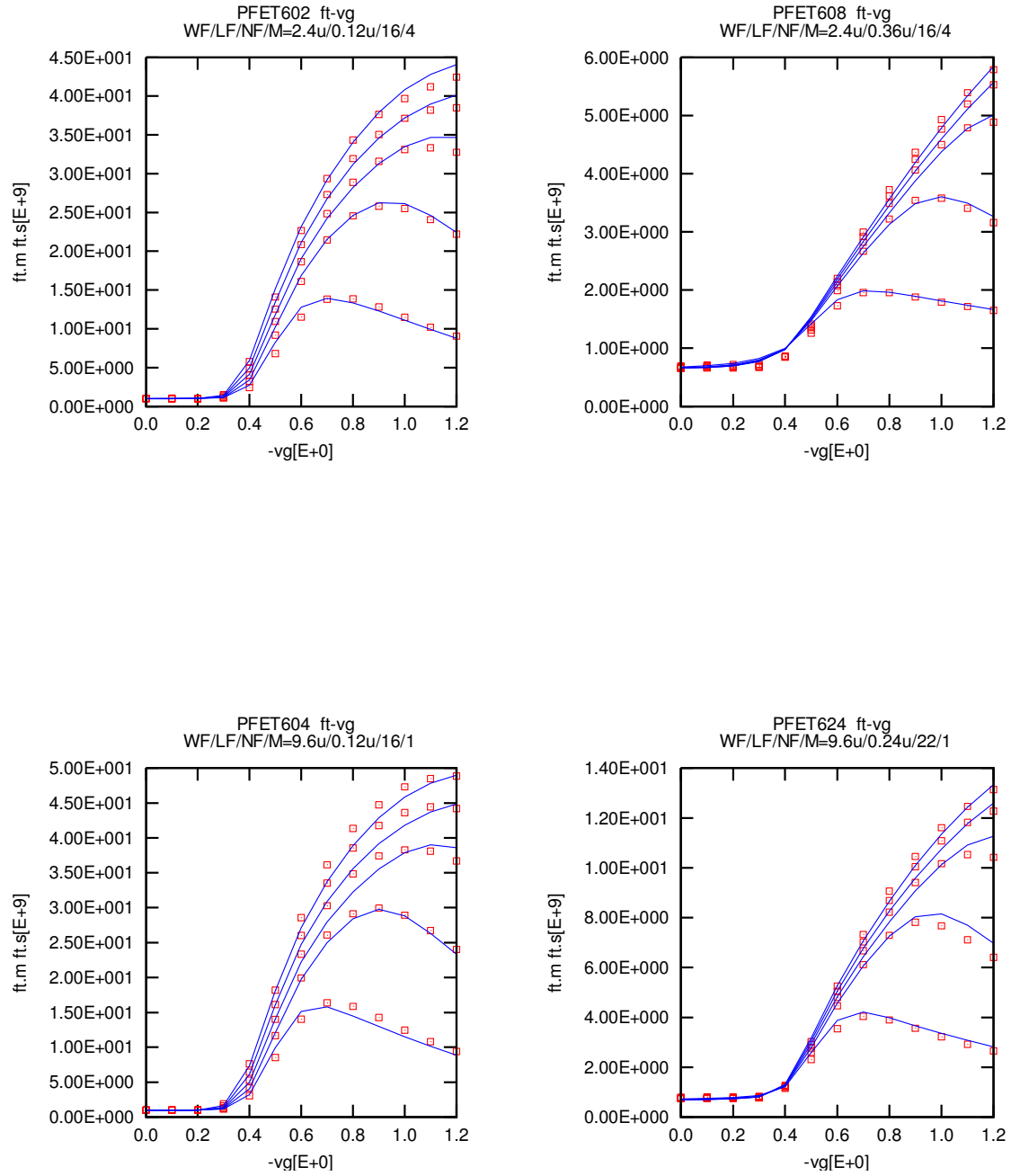


Figure 2-21 fT-Vg (25°C)

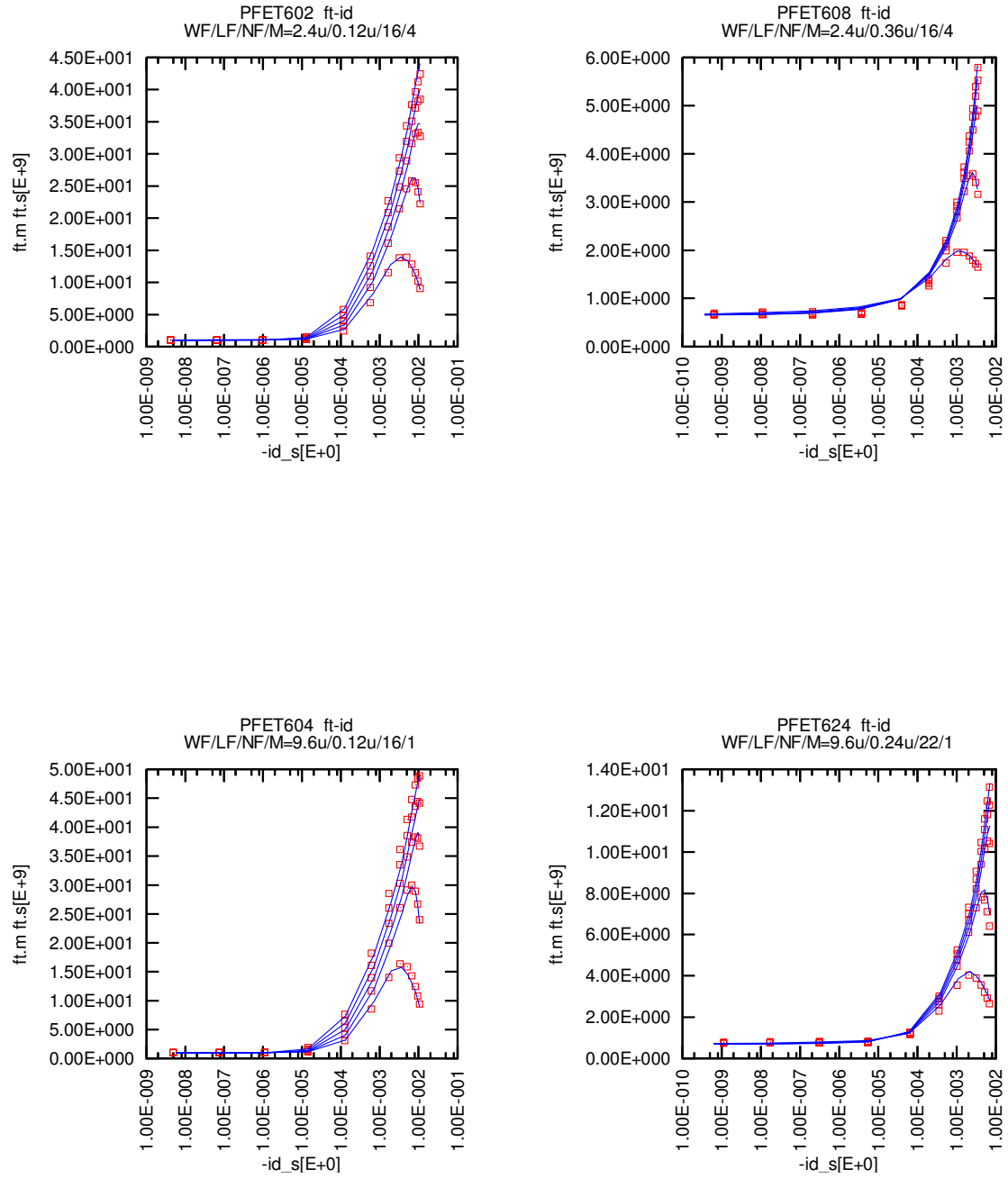
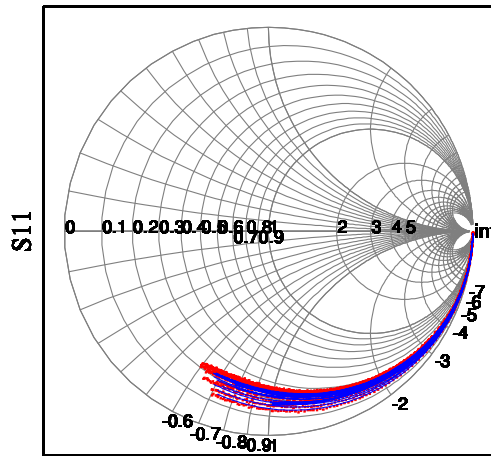
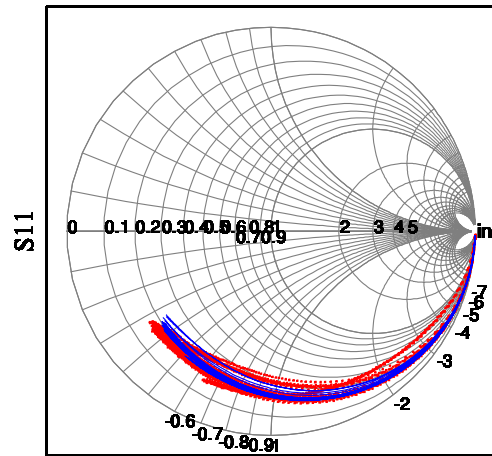


Figure 2-22 f_{T-id} (25°C)

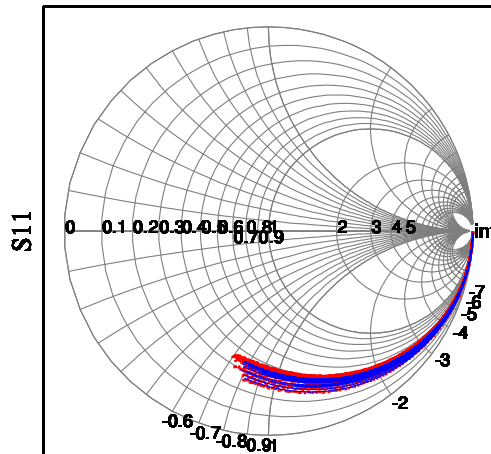
PFET602 S11 WF/LF/NF/M=2.4u/0.12u/16/4



PFET608 S11 WF/LF/NF/M=2.4u/0.36u/16/4



PFET604 S11 WF/LF/NF/M=9.6u/0.12u/16/1



PFET624 S11 WF/LF/NF/M=9.6u/0.24u/22/1

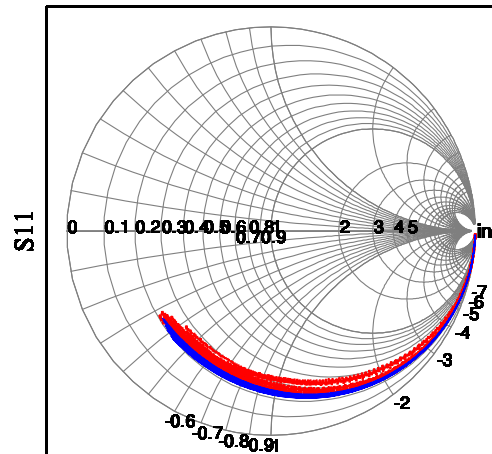


Figure 2-23 S11 (25°C)

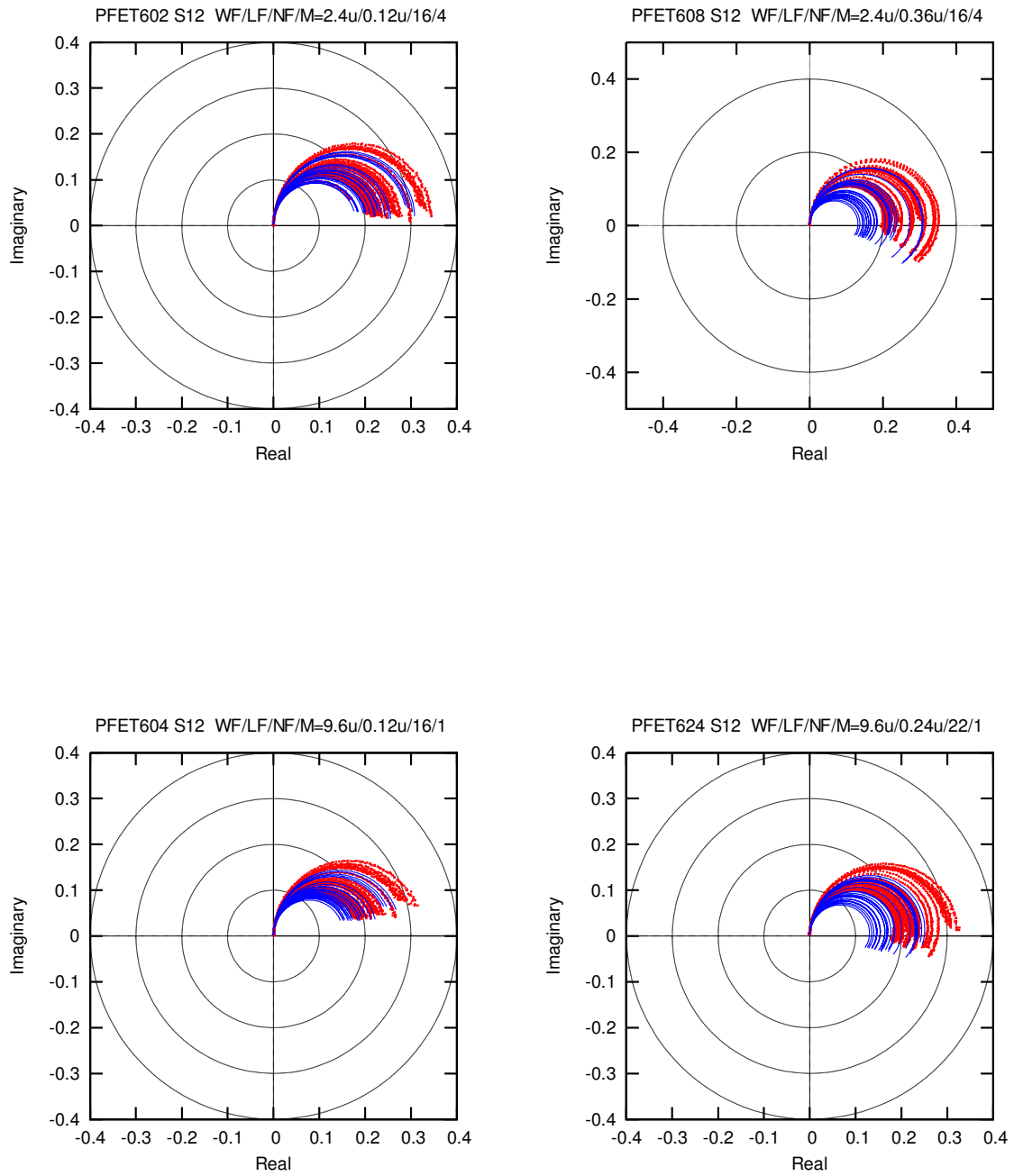


Figure 2-24 S12 (25°C)

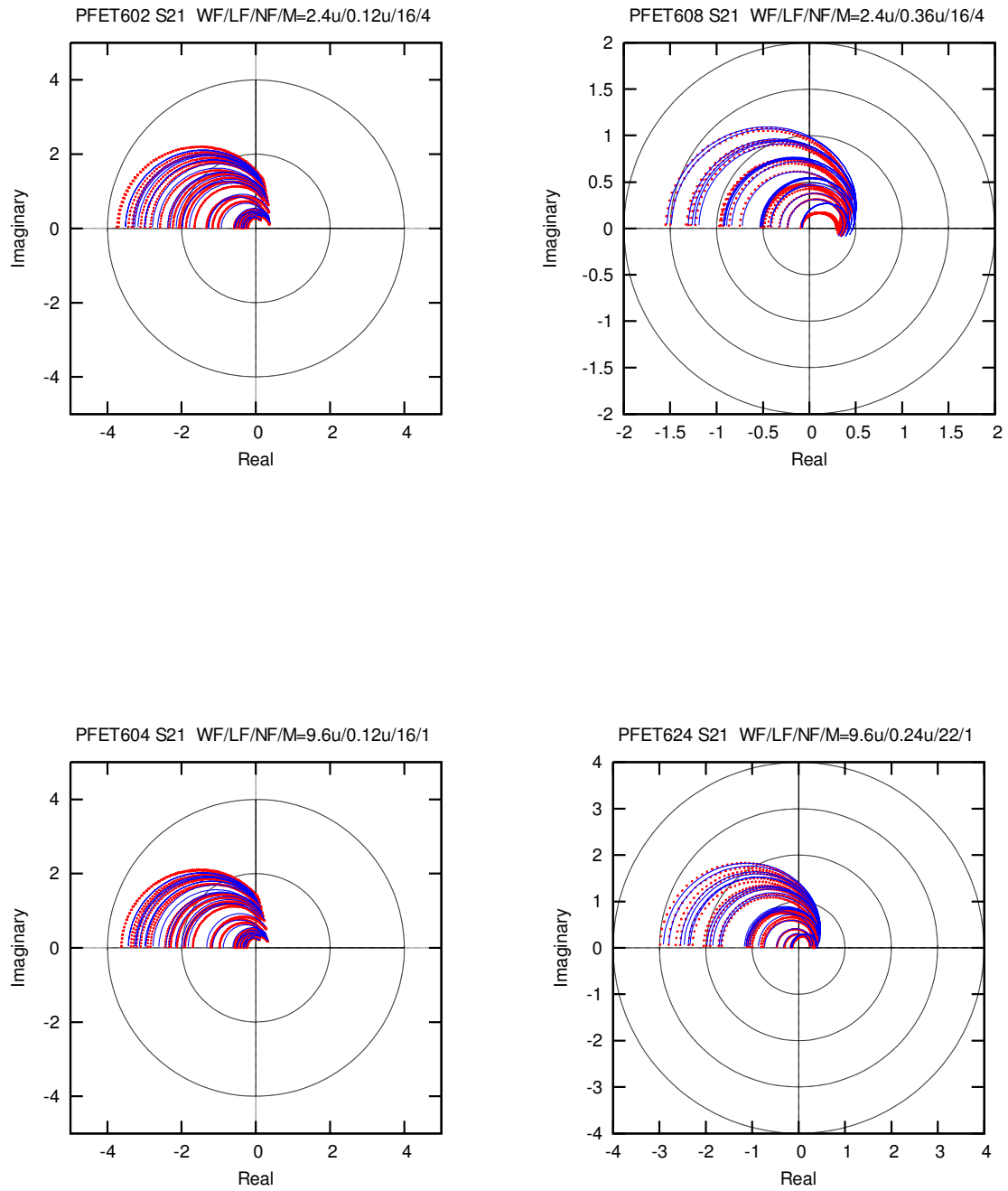
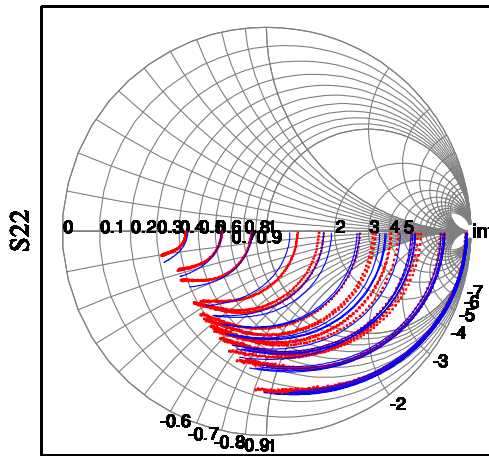
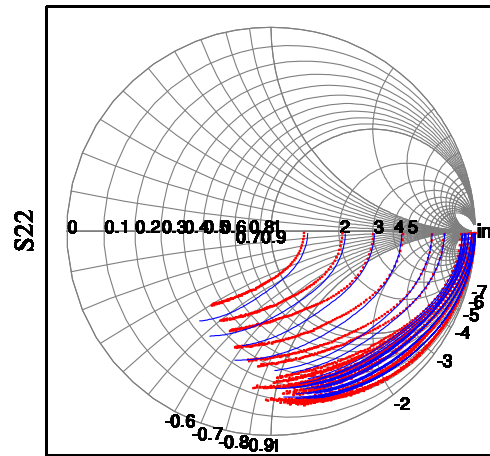


Figure 2-25 S21 (25°C)

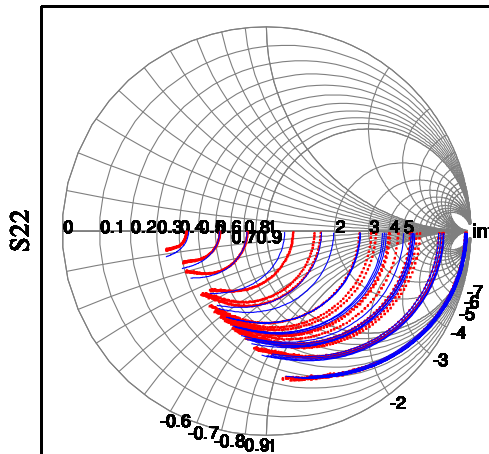
PFET602 S22 WF/LF/NF/M=2.4u/0.12u/16/4



PFET608 S22 WF/LF/NF/M=2.4u/0.36u/16/4



PFET604 S22 WF/LF/NF/M=9.6u/0.12u/16/1



PFET624 S22 WF/LF/NF/M=9.6u/0.24u/22/1

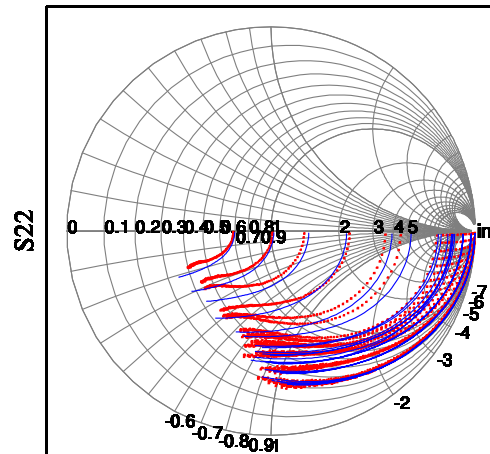


Figure 2-26 S22 (25°C)

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	n_12_sprf & n_12bpw_sprf warning: Acde="2.83e-01" is unusually small.	A small value of parameter acde is necessary to get good CV fitting but would cause a harmless warning in HSPICE.

3.2 SPECTRE Warning Message

Table 3-2 SPECTRE warning message

NO.	Warning Message	Explanation
1.	n_12_sprf & n_12bpw_sprf warning: Acde="2.83e-01" is unusually small.	A small value of parameter acde is necessary to get good CV fitting but would cause a harmless warning in ELDO.

3.3 ELDO Warning Message

Table 3-3 ELDO warning message

NO.	Warning Message	Explanation
1.	n_12_sprf & n_12bpw_sprf warning: Acde="2.83e-01" is unusually small.	A small value of parameter acde is necessary to get good CV fitting but would cause a harmless warning in SPECTRE.